Supplementary material to "Hypothesis testing for estimable parameters with missing data: available cases or complete cases?"

M. Romero-Madroñal*[1], M.D. Jiménez-Gamero[1], and B. Milošević[2]

[1]Department of Statistics and Operations Research, Universidad de Sevilla, Spain , Instituto de Matemáticas de la Universidad de Sevilla, Spain

[2]Faculty of Mathematics, University of Belgrade, Faculty of Mathematics, Serbia

This supplementary material contains the proofs of all the results stated in the paper (Section S1), the classical results on U-statistics used in them (Section S2), the description of the Gaussian copula missingness mechanism (Section S3) and the average sample sizes of the simulation study of Subsection 8.1 of the paper (Section S4). Sections, equations and tables of this supplement are numbered with the prefix "S"; lemmas and remarks continue the numbering of the paper. References without the prefix "S" refer to the paper.

S1 Proofs

A sequence of random variables is $o_{\mathbb{P}}(1)$ if it converges to 0, and is $O_{\mathbb{P}}(1)$ if it is bounded in probability. Let $\|\cdot\|$ denote the Euclidean norm. As said in Section 2, we are considering the setting under Assumption A4. For brevity in notation, we remove the indicator of this event in all proofs.

Remark 9. *Since* $R_j^{(r)} \in \{0,1\}$,

$$\binom{N^{(r)}}{m_r} = \sum_{1\le j_1<\cdots<j_{m_r}\le n} \prod_{a=1}^{m_r} R_{j_a}^{(r)},$$

*Corresponding author. Email address: mrmadronal@us.es

because the product equals one if and only if all selected observations are available.

Proof of Proposition 1. From (9),

$$\mathbb{E}\big(\widehat{\eta}_{AC}^{(r)} \,\big|\, R_1^{(r)},\ldots,R_n^{(r)}\big) = \binom{N^{(r)}}{m_r}^{-1} \sum_{1\le j_1<\cdots<j_{m_r}\le n} \prod_{a=1}^{m_r} R_{j_a}^{(r)}\, \mathbb{E}\Big\{h^{(r)}\big(X_{j_1}^{A_r},\ldots,X_{j_{m_r}}^{A_r}\big) \,\Big|\, R_1^{(r)},\ldots,R_n^{(r)}\Big\}.$$

By Assumption A3, each conditional expectation equals $\mathbb{E}\big\{h^{(r)}\big(X_{j_1}^{A_r},\ldots,X_{j_{m_r}}^{A_r}\big)\big\} = \eta^{(r)}$, and, from Remark 9, the remaining sum equals $\binom{N^{(r)}}{m_r}$. □

Proof of Proposition 2. From Assumption A3 we have that

$$N^{(r)} = \sum_{j=1}^{n} R_j^{(r)} \sim \mathrm{Bin}\big(n, \pi_n^{(r)}\big), \tag{S1}$$

with mean $n\pi_n^{(r)}$ and variance $n\pi_n^{(r)}\big(1-\pi_n^{(r)}\big) \le n\pi_n^{(r)}$. If $n\pi_n^{(r)} \ge 2m_r$, then

$$n\pi_n^{(r)} - (m_r - 1) \ge n\pi_n^{(r)} - m_r \ge \frac{n\pi_n^{(r)}}{2}.$$

Therefore,

$$\big(E_n^{(r)}\big)^c = \big\{N^{(r)} < m_r\big\} = \big\{N^{(r)} \le m_r - 1\big\} \subseteq \left\{\big|N^{(r)} - n\pi_n^{(r)}\big| \ge \frac{n\pi_n^{(r)}}{2}\right\}.$$

From Chebyshev's inequality,

$$\mathbb{P}\big\{(E_n^{(r)})^c\big\} \le \mathbb{P}\left(\big|N^{(r)} - n\pi_n^{(r)}\big| \ge \frac{n\pi_n^{(r)}}{2}\right) \le \frac{\mathbb{V}(N^{(r)})}{\Big(n\pi_n^{(r)}/2\Big)^2} \le \frac{4}{n\pi_n^{(r)}}.$$

Finally, since

$$E_n^c = \left(\bigcap_{r=1}^{q} E_n^{(r)}\right)^c = \bigcup_{r=1}^{q} \big(E_n^{(r)}\big)^c,$$

the union bound gives

$$\mathbb{P}(E_n^c) \le \sum_{r=1}^{q} \mathbb{P}\big\{(E_n^{(r)})^c\big\} \le \sum_{r=1}^{q} \frac{4}{n\pi_n^{(r)}}.$$

Recalling A4 the result holds. □

Proof of Proposition 3. (a) From (S1), we have that $\mathbb{E}(N^{(r)2}) = \mathbb{E}^2(N^{(r)}) + \mathbb{V}(N^{(r)}) = (n\pi_n^{(r)})^2 + n\pi_n^{(r)}(1-\pi_n^{(r)}) \to 0$, and thus, by Markov inequality, we get that, $\mathbb{P}(E_n^{(r)}) \to 0$. As $\mathbb{P}(E_n) \le \mathbb{P}(E_n^{(r)})$, the result follows.

(b) From (S1), it follows that $N^{(r)} \xrightarrow{\mathcal{L}} Y$, and hence the result. □

Before proving Theorem 1, we give some preliminary results. First, we observe the following: the AC estimator defined in (9) can be witten as

$$\widehat{\eta}_{AC}^{(r)} = \frac{A_n^{(r)}}{B_n^{(r)}}.$$

Both $A_n^{(r)}$ and $B_n^{(r)}$ are U-statistics of degree m_r in the augmented observations Z_j. Their kernels are, respectively,

$$\psi_A^{(r)}(Z_1, \ldots, Z_{m_r}) = h^{(r)}\big(X_1^{A_r}, \ldots, X_{m_r}^{A_r}\big) \prod_{a=1}^{m_r} R_a^{(r)} \tag{S2}$$

and

$$\psi_B^{(r)}(Z_1, \ldots, Z_{m_r}) = \prod_{a=1}^{m_r} R_a^{(r)}. \tag{S3}$$

From Corollary 1, we know that $\mathbb{E}(\widehat{\eta}_{AC}^{(r)}) = \eta^{(r)}$, and then, the centered U-statistic $\widehat{\eta}_{AC}^{(r)} - \eta^{(r)}$ can be expressed as

$$\widehat{\eta}_{AC}^{(r)} - \eta^{(r)} = \frac{A_n^{(r)} - \eta^{(r)} B_n^{(r)}}{B_n^{(r)}} := \frac{U_n^{(r)}}{B_n^{(r)}}, \tag{S4}$$

where the kernel of the numerator $U_n^{(r)}$ is given by

$$\psi^{(r)}(Z_1, \ldots, Z_{m_r}) = \psi_A^{(r)}(Z_1, \ldots, Z_{m_r}) - \eta^{(r)}\, \psi_B^{(r)}(Z_1, \ldots, Z_{m_r}). \tag{S5}$$

Lemma 2. *For integers $1 \le c \le m$, and $n \ge 2m$, let*

$$\gamma_{c,n} := \frac{n\binom{m}{c}\binom{n-m}{m-c}}{\binom{n}{m}}.$$

Then,

1. $\gamma_{1,n} = m^2 \prod_{i=0}^{m-2} \frac{n-m-i}{n-1-i} \;\le\; m^2$.
2. $\gamma_{c,n} \le K_m\, n^{1-c}$, $2 \le c \le m$, *with* $K_m = 4^m m!$.

Proof. 1. Writing out the binomial coefficients,

$$\frac{\binom{n-m}{m-1}}{\binom{n}{m}} = \frac{(n-m)!}{(m-1)!\,(n-2m+1)!} \frac{m!\,(n-m)!}{n!} = m \frac{(n-m)!}{(n-2m+1)!} \frac{(n-m)!}{n!}.$$

Now

$$\frac{(n-m)!}{(n-2m+1)!} = (n-m)(n-m-1)\cdots(n-2m+2) = \prod_{i=0}^{m-2}(n-m-i),$$

and

$$\frac{(n-m)!}{n!} = \frac{1}{n(n-1)\cdots(n-m+1)} = \frac{1}{\prod_{i=0}^{m-1}(n-i)}.$$

Hence

$$\gamma_{1,n} = nm^2\frac{\prod_{i=0}^{m-2}(n-m-i)}{\prod_{i=0}^{m-1}(n-i)} = m^2\,\frac{\prod_{i=0}^{m-2}(n-m-i)}{\prod_{i=0}^{m-2}(n-1-i)},$$

where the factor n canceled the $i=0$ term of the denominator and the remaining product was reindexed $\prod_{i=1}^{m-1}(n-i) = \prod_{i=0}^{m-2}(n-1-i)$. Each factor $(n-m-i)/(n-1-i) = 1-(m-1)/(n-1-i)$ lies in $(0,1]$ because $m \geq 1$, which gives $\gamma_{1,n} \leq m^2$, and tends to 1 as $n \to \infty$, which gives $\gamma_{1,n} \to m^2$ for $m = 1$ the product is empty and $\gamma_{1,n} = 1 = m^2$ exactly.

2. We first bound each binomial coefficient separately. Since $\binom{a}{b} \leq a^b/b!$, we obtain

$$\binom{n-m}{m-c} \leq \frac{(n-m)^{m-c}}{(m-c)!} \leq \frac{n^{m-c}}{(m-c)!}.$$

For the denominator, note that

$$\binom{n}{m} = \frac{\prod_{i=0}^{m-1}(n-i)}{m!} \geq \frac{(n-m+1)^m}{m!}.$$

Moreover, if $n \geq 2m$, then $n-m+1 \geq n/2$, and hence

$$\binom{n}{m} \geq \frac{(n/2)^m}{m!}, \qquad \text{or equivalently} \qquad \binom{n}{m}^{-1} \leq m!\,2^m\,n^{-m}.$$

Substituting these bounds into the definition of $\gamma_{c,n}$ gives

$$\gamma_{c,n} \leq n\binom{m}{c}\frac{n^{m-c}}{(m-c)!}\cdot m!\,2^m\,n^{-m} = \frac{\binom{m}{c}\,m!\,2^m}{(m-c)!}\,n^{1-c}.$$

Finally, using the elementary bounds $\binom{m}{c} \leq 2^m$, $(m-c)! \geq 1$, we conclude the result.

□

Lemma 3. *Suppose that Assumptions A3-A4 hold. Then*

$$\frac{\left(\pi_n^{(r)}\right)^m}{B_n^{(r)}} \xrightarrow{\mathbb{P}} 1, \quad 1 \le r \le q. \tag{S6}$$

Proof. Fix $1 \le r \le q$. For simplicity in notation write

$$m := m_r, \qquad \pi_n := \pi_n^{(r)}, \quad R_j := R_j^{(r)}, \quad h_1 := h_1^{(r)}. \tag{S7}$$

From Remark 9, $B_n^{(r)} = \binom{N^{(r)}}{m}\binom{n}{m}^{-1} \ge \binom{n}{m}^{-1} > 0$. Assumption A3 implies that $R_1, \ldots, R_m$ are i.i.d. Bernoulli(π_n) random variables, so we have that $\mathbb{E}\big(B_n^{(r)}\big) = \mathbb{E}\big\{\psi_B^{(r)}(Z_1, \ldots, Z_m)\big\} = \mathbb{E}\big(\prod_{a=1}^m R_a\big) = \pi_n^m$. Moreover, the conditional expectations (S53) associated with the kernel $\psi_B^{(r)}$ in (S3) are

$$\mathbb{E}\Big(\prod_{a=1}^m R_a \,\Big|\, Z_1, \ldots, Z_c\Big) = \pi_n^{m-c} \prod_{a=1}^c R_a, \qquad 1 \le c \le m,$$

and, since $\prod_{a=1}^c R_a$ is a Bernoulli(π_n^c) random variable, its variance is $\pi_n^{2(m-c)}\,\pi_n^c\big(1-\pi_n^c\big) = \pi_n^{2m-c}\big(1-\pi_n^c\big)$, $1 \le c \le m$. Therefore, from (S58) and the definition of $\gamma_{c,n}$ in Lemma 2,

$$\mathbb{V}\big(B_n^{(r)}\big) = \binom{n}{m}^{-1} \sum_{c=1}^m \binom{m}{c}\binom{n-m}{m-c} \pi_n^{2m-c}\,\big(1-\pi_n^c\big) = \frac{1}{n}\sum_{c=1}^m \gamma_{c,n}\,\pi_n^{2m-c}\,\big(1-\pi_n^c\big).$$

Let n be large enough so that $n \ge 2m$ and $n\pi_n \ge 2$, which is possible because $n\pi_n \to \infty$ from Assumption A4. Bounding $1 - \pi_n^c \le 1$ and using Lemma 2, the summand with $c = 1$ is at most $m^2\,\pi_n^{2m-1}$, while, for $2 \le c \le m$, $\gamma_{c,n}\,\pi_n^{2m-c} \le K_m\,n^{1-c}\,\pi_n^{2m-c} = K_m\,\pi_n^{2m-1}\,(n\pi_n)^{1-c}$ and $\sum_{c=2}^m (n\pi_n)^{1-c} \le \sum_{k=1}^\infty 2^{-k} = 1$. Hence,

$$\mathbb{V}\big(B_n^{(r)}\big) \le \frac{C_m\,\pi_n^{2m-1}}{n}, \tag{S8}$$

where $C_m = m^2 + K_m$. Chebyshev's inequality and (S8) then give, for every $\varepsilon > 0$,

$$\mathbb{P}\Big(\big|B_n^{(r)}/\pi_n^m - 1\big| > \varepsilon\Big) = \mathbb{P}\Big(\big|B_n^{(r)} - \pi_n^m\big| > \varepsilon\,\pi_n^m\Big) \le \frac{\mathbb{V}\big(B_n^{(r)}\big)}{\varepsilon^2\,\pi_n^{2m}} \le \frac{C_m}{\varepsilon^2\,n\pi_n}, \tag{S9}$$

and, since $n\pi_n \to \infty$, this proves $B_n^{(r)}/\pi_n^m \xrightarrow{\mathbb{P}} 1$. Because $B_n^{(r)}$ is positive, as noted at the beginning of the proof, and the map $x \mapsto 1/x$ is continuous at $x = 1$, the Continuous Mapping Theorem and $B_n^{(r)}/\pi_n^m \xrightarrow{\mathbb{P}} 1$ yield (S6). □

Lemma 4. *Under Assumptions A1, A3 and A4,*

$$\mathcal{D}_n\left(\widehat{\eta}_{AC}-\eta\right)=\frac{1}{\sqrt{n}}\sum_{j=1}^{n}\xi_{n,j}\ +\ r_n,\qquad r_n\ \xrightarrow{\mathbb{P}}\ 0, \tag{S10}$$

with

$$\xi_{n,j}=\left(\xi_{n,j}^{(1)},\ldots,\xi_{n,j}^{(q)}\right):=\left(\frac{m_1}{\left(\pi_n^{(1)}\right)^{1/2}}\,R_j^{(1)}\,\tilde{h}_1^{(1)}\big(X_j^{A_1}\big),\ \ldots,\ \frac{m_q}{\left(\pi_n^{(q)}\right)^{1/2}}\,R_j^{(q)}\,\tilde{h}_1^{(q)}\big(X_j^{A_q}\big)\right)^{\top},\quad 1\le j\le n, \tag{S11}$$

where $\tilde{h}_1^{(r)}$, $1\le r\le q$, denotes the centered first-order projection of $h^{(r)}$, as defined in (S54).

Proof. Let $r_n=(r_n^{(1)},\ldots,r_n^{(q)})^{\top}$. Since $\|r_n\|\le\sum_{r=1}^{q}\big|r_n^{(r)}\big|$ and, for every $\varepsilon>0$, $\mathbb{P}\big(\sum_{r=1}^{q}|r_n^{(r)}|>\varepsilon\big)\le\sum_{r=1}^{q}\mathbb{P}\big(|r_n^{(r)}|>\varepsilon/q\big)$, it suffices to prove $r_n^{(r)}\xrightarrow{\mathbb{P}}0$ for each fixed r. Fix $1\le r\le q$ and use the abbreviations in (S7). Write $U_n:=U_n^{(r)}$ as defined in (S4), and let ψ_c, $1\le c\le m$, denote the conditional expectations (S53) associated with the kernel ψ in (S5) and the i.i.d. sample $Z_1,\ldots,Z_n$. Likewise, let $\tilde{h}_c$ and σ_c^2, $1\le c\le m$, be as in (S54) and (S55), applied to the kernel $h^{(r)}$, respectively. From Assumption A1, $\sigma_m^2=\mathbb{V}\big\{h^{(r)}\big(X_1^{A_r},\ldots,X_m^{A_r}\big)\big\}<\infty$.

Step 1: projections of the kernel ψ. From Assumption A3, $R_1,\ldots,R_m$ are i.i.d. Bernoulli(π_n) random variables and $(R_1,\ldots,R_m)$ is independent of $(X_1,\ldots,X_m)$. Hence, taking conditional expectations in $\psi=\big\{h^{(r)}\big(X_1^{A_r},\ldots,X_m^{A_r}\big)-\eta^{(r)}\big\}\prod_{a=1}^{m}R_a$, and using that $\prod_{a=c+1}^{m}R_a$ is independent of $\big(R_1,\ldots,R_c,X_1,\ldots,X_m\big)$ and has mean π_n^{m-c}, we have that

$$\psi_c(Z_1,\ldots,Z_c)=\pi_n^{m-c}\prod_{a=1}^{c}R_a\,\tilde{h}_c\big(X_1^{A_r},\ldots,X_c^{A_r}\big),\qquad 1\le c\le m. \tag{S12}$$

Since $R_a^2=R_a$, $\mathbb{E}\big\{\tilde{h}_c\big(X_1^{A_r},\ldots,X_c^{A_r}\big)\big\}=0$ and the fact that $(R_1,\ldots,R_c)$ is independent of $(X_1,\ldots,X_c)$, (S12) gives $\mathbb{E}\{\psi_c(Z_1,\ldots,Z_c)\}=0$ and

$$s_c^2:=\mathbb{V}\{\psi_c(Z_1,\ldots,Z_c)\}=\pi_n^{2(m-c)}\,\pi_n^{c}\,\sigma_c^2=\pi_n^{2m-c}\,\sigma_c^2,\qquad 1\le c\le m. \tag{S13}$$

In particular, $\mathbb{E}(U_n)=\mathbb{E}(\psi)=0$ and, from (S57), $s_c^2\le\pi_n^{2m-c}\,\sigma_m^2$, $1\le c\le m$.

Step 2: decomposition of $r_n^{(r)}$. Let $Y_n:=\frac{m}{\sqrt{n}}\sum_{j=1}^{n}\psi_1(Z_j)$ and $T_n:=\sqrt{n}\,U_n$. From (S12) with $c=1$, $R_j\,\tilde{h}_1\big(X_j^{A_r}\big)=\pi_n^{1-m}\,\psi_1(Z_j)$, $1\le j\le n$, so that

$$\frac{1}{\sqrt{n}}\sum_{j=1}^{n}\xi_{n,j}^{(r)}=\frac{m}{\sqrt{n\pi_n}}\sum_{j=1}^{n}R_j\,\tilde{h}_1\big(X_j^{A_r}\big)=\pi_n^{1/2}\,\frac{Y_n}{\pi_n^{m}}. \tag{S14}$$

From (S4), it follows that $\widehat{\eta}_{AC}^{(r)} - \eta^{(r)} = U_n / B_n^{(r)}$, and then

$$\sqrt{n\pi_n}\left(\widehat{\eta}_{AC}^{(r)} - \eta^{(r)}\right) = \pi_n^{1/2}\sqrt{n}\left(\widehat{\eta}_{AC}^{(r)} - \eta^{(r)}\right) = \pi_n^{1/2}\frac{T_n}{B_n^{(r)}}. \tag{S15}$$

Next, note that

$$\frac{T_n}{B_n^{(r)}} - \frac{Y_n}{\pi_n^m} = \frac{T_n - Y_n}{\pi_n^m} + T_n\Big(\frac{1}{B_n^{(r)}} - \frac{1}{\pi_n^m}\Big) = \frac{T_n - Y_n}{\pi_n^m} + \frac{T_n}{\pi_n^m}\Big(\frac{\pi_n^m}{B_n^{(r)}} - 1\Big), \tag{S16}$$

and then, recalling the definition of r_n in (S10), from (S14), (S15) and (S16), we obtain

$$r_n^{(r)} = \pi_n^{1/2}\frac{T_n - Y_n}{\pi_n^m} + \pi_n^{1/2}\frac{T_n}{\pi_n^m}\Big(\frac{\pi_n^m}{B_n^{(r)}} - 1\Big). \tag{S17}$$

We show that each term in (S17) converges to zero in probability.

Step 3: first term on the right-hand side of (S17). From Step 1, $\mathbb{E}(T_n) = \sqrt{n}\,\mathbb{E}(U_n) = 0$ and $\psi_1(Z_1), \ldots, \psi_1(Z_n)$ are i.i.d. with zero mean, so $\mathbb{E}(Y_n) = 0$ and, from (S13) with $c = 1$, $\mathbb{V}(Y_n) = m^2\,\mathbb{V}\{\psi_1(Z_1)\} = m^2\,\pi_n^{2m-1}\,\sigma_1^2$. Therefore,

$$\mathbb{E}\big\{(T_n - Y_n)^2\big\} = \mathbb{V}(T_n - Y_n) = \mathbb{V}(T_n) - 2\,\mathbb{C}\mathrm{ov}(T_n, Y_n) + \mathbb{V}(Y_n). \tag{S18}$$

We now compute the first two terms in (S18). First, applying (S58) to U_n, whose kernel is ψ, and using (S13) together with the definition of $\gamma_{c,n}$ in Lemma 2,

$$\mathbb{V}(T_n) = n\,\mathbb{V}(U_n) = \sum_{c=1}^{m} n\binom{n}{m}^{-1}\binom{m}{c}\binom{n-m}{m-c}\mathbb{V}\{\psi_c(Z_1, \ldots, Z_c)\} = \sum_{c=1}^{m}\gamma_{c,n}\,\pi_n^{2m-c}\,\sigma_c^2.$$

Second, if $j \notin \{i_1, \ldots, i_m\}$, then $\mathbb{C}\mathrm{ov}\big\{\psi(Z_{i_1}, \ldots, Z_{i_m}), \psi_1(Z_j)\big\} = 0$, because the two variables are independent, while, if $j \in \{i_1, \ldots, i_m\}$, the index sets $\{i_1, \ldots, i_m\}$ and $\{j\}$ share exactly one element, so (S56), applied to the kernel ψ, gives $\mathbb{C}\mathrm{ov}\big\{\psi(Z_{i_1}, \ldots, Z_{i_m}), \psi_1(Z_j)\big\} = \mathbb{V}\{\psi_1(Z_1)\} = \pi_n^{2m-1}\,\sigma_1^2$. Since each set $\{i_1, \ldots, i_m\}$ contains exactly m indices j,

$$\begin{aligned}\mathbb{C}\mathrm{ov}(T_n, Y_n) &= m\binom{n}{m}^{-1}\sum_{1\le i_1<\cdots<i_m\le n}\;\sum_{j=1}^{n}\mathbb{C}\mathrm{ov}\big\{\psi(Z_{i_1}, \ldots, Z_{i_m}), \psi_1(Z_j)\big\} \\ &= m\binom{n}{m}^{-1}\binom{n}{m}\,m\,\pi_n^{2m-1}\,\sigma_1^2 = \mathbb{V}(Y_n).\end{aligned}$$

Substituting the three terms into (S18), the $c = 1$ summand of $\mathbb{V}(T_n)$ partially cancels, and we obtain

$$\mathbb{E}\big\{(T_n - Y_n)^2\big\} = \big(\gamma_{1,n} - m^2\big)\,\pi_n^{2m-1}\,\sigma_1^2 + \sum_{c=2}^{m}\gamma_{c,n}\,\pi_n^{2m-c}\,\sigma_c^2. \tag{S19}$$

Let n be large enough so that $n \geq 2m$ and $n\pi_n \geq 2$, which is possible because $n\pi_n \to \infty$ from Assumption A4. From Lemma 2, $\gamma_{1,n} \leq m^2$, so the first summand in (S19) is nonpositive. For the remaining ones, Lemma 2 and (S57) give

$$\gamma_{c,n}\,\pi_n^{2m-c}\,\sigma_c^2 \leq K_m\,n^{1-c}\,\pi_n^{2m-c}\,\sigma_m^2 = K_m\,\sigma_m^2\,n\,\pi_n^{2m}\,(n\pi_n)^{-c}, \qquad 2 \leq c \leq m,$$

and, since $n\pi_n \geq 2$,

$$\sum_{c=2}^{m}(n\pi_n)^{-c} \leq (n\pi_n)^{-2}\sum_{k=0}^{\infty}(n\pi_n)^{-k} = \frac{(n\pi_n)^{-2}}{1-(n\pi_n)^{-1}} \leq 2\,(n\pi_n)^{-2}.$$

Combining the last three displays, $\mathbb{E}\big\{(T_n - Y_n)^2\big\} \leq 2K_m\,\sigma_m^2\,\pi_n^{2m-2}/n$ and, multiplying by $\big(\pi_n^{1/2}/\pi_n^m\big)^2 = \pi_n^{1-2m}$,

$$\mathbb{E}\left[\left\{\pi_n^{1/2}\,\frac{T_n - Y_n}{\pi_n^m}\right\}^2\right] \leq \frac{2K_m\,\sigma_m^2}{n\pi_n} \longrightarrow 0,$$

by Assumption A4. Finally, from Markov's inequality, for every $\varepsilon > 0$, $\mathbb{P}\big(\pi_n^{1/2}\,|T_n - Y_n|/\pi_n^m > \varepsilon\big) \leq \varepsilon^{-2}\,\mathbb{E}\big[\big\{\pi_n^{1/2}\,(T_n - Y_n)/\pi_n^m\big\}^2\big] \to 0$, that is, $\pi_n^{1/2}\,(T_n - Y_n)/\pi_n^m \xrightarrow{\mathbb{P}} 0$.

Step 4: second term on the right-hand side of (S17). The second term in (S17) is the product of $\pi_n^{1/2}\,T_n/\pi_n^m$, which is bounded in mean square,

$$\mathbb{E}\left[\left\{\pi_n^{1/2}\,\frac{T_n}{\pi_n^m}\right\}^2\right] = \pi_n^{1-2m}\,\mathbb{V}(T_n) = \sum_{c=1}^{m}\gamma_{c,n}\,\pi_n^{1-c}\,\sigma_c^2 \leq m^2\sigma_m^2 + K_m\,\sigma_m^2\sum_{c=2}^{m}(n\pi_n)^{1-c} \leq \big(m^2 + K_m\big)\,\sigma_m^2 < \infty,$$

and $\pi_n^m/B_n^{(r)} - 1$, which converges to zero in probability from Lemma 3. Hence the product is $o_{\mathbb{P}}(1)$, which completes the proof. □

Proof of Theorem 1. From Lemma 4, $\mathcal{D}_n\big(\widehat{\eta}_{AC} - \eta\big) = W_n + r_n$ with $r_n \xrightarrow{\mathbb{P}} 0$, where

$$W_n = \frac{1}{\sqrt{n}}\sum_{j=1}^{n}\xi_{n,j}, \tag{S20}$$

and $\xi_{n,j}$ is defined in (S11). Hence, from Slutsky's theorem, it suffices to prove that $W_n \xrightarrow{\mathcal{L}} \mathcal{N}_q\big(0, V \circ \varrho\big)$.

Step 1: mean and covariance of $\xi_{n,1}$. From Assumption A3, $Z_1, \ldots, Z_n$ are i.i.d. and, since, for each n, $\xi_{n,j}$ is a fixed measurable function of Z_j, the random vectors $\xi_{n,1}, \ldots, \xi_{n,n}$ are i.i.d. From (8), $\big(R_1^{(1)}, \ldots, R_1^{(q)}\big)$ is a function of $\big(M_1^{(1)}, \ldots, M_1^{(p)}\big)$, which, from (7), is independent of X_1. Hence, from (S11) and $\mathbb{E}\big\{\tilde{h}_1^{(r)}\big(X_1^{A_r}\big)\big\} = 0$, $1 \leq r \leq q$, which follows

from (S54), we obtain $\mathbb{E}\big(\xi_{n,1}\big) = 0$ and

$$\mathbb{C}\mathrm{ov}\big(\xi_{n,1}^{(r)}, \xi_{n,1}^{(s)}\big) = \frac{m_r m_s}{\sqrt{\pi_n^{(r)}\pi_n^{(s)}}}\,\mathbb{E}\Big\{R_1^{(r)} R_1^{(s)}\,\tilde{h}_1^{(r)}\big(X_1^{A_r}\big)\,\tilde{h}_1^{(s)}\big(X_1^{A_s}\big)\Big\} = \frac{m_r m_s\,\pi_n^{(r,s)}}{\sqrt{\pi_n^{(r)}\pi_n^{(s)}}}\,\Gamma_{rs} = m_r m_s\,\Gamma_{rs}\,\varrho_n^{(r,s)}, \tag{S21}$$

$1 \le r, s \le q$, where the second equality uses the independence stated above together with $\mathbb{E}\big(R_1^{(r)} R_1^{(s)}\big) = \pi_n^{(r,s)}$ (see (11)), $\mathbb{E}\big\{\tilde{h}_1^{(r)}\big(X_1^{A_r}\big)\tilde{h}_1^{(s)}\big(X_1^{A_s}\big)\big\} = \mathbb{C}\mathrm{ov}\big\{h_1^{(r)}\big(X_1^{A_r}\big), h_1^{(s)}\big(X_1^{A_s}\big)\big\} = \Gamma_{rs}$ (see (4)) and (12). In matrix form, $\mathbb{V}\big(\xi_{n,1}\big) = V \circ \varrho_n$, with $\varrho_n := \big(\varrho_n^{(r,s)}\big)_{1\le r,s\le q}$ and V as in (5), and, from Assumption A5, $V \circ \varrho_n \to V \circ \varrho$ entrywise. In particular, $V \circ \varrho$ is a limit of covariance matrices and hence positive semidefinite, so the limit law $\mathcal{N}_q\big(0, V \circ \varrho\big)$ is well defined.

Step 2: Cramér–Wold reduction. From the Cramér–Wold device, it suffices to show that, for each fixed $\lambda = (\lambda_1, \ldots, \lambda_q)^\top \in \mathbb{R}^q$,

$$\lambda^\top W_n \;\xrightarrow{\mathcal{L}}\; \mathcal{N}\big(0, \sigma_\lambda^2\big), \tag{S22}$$

where $\sigma_\lambda^2 = \lambda^\top\big(V \circ \varrho\big)\lambda < \infty$. Fix $\lambda \in \mathbb{R}^q$. From Step 1, $\lambda^\top W_n$ has zero mean and, using that the summands are i.i.d., its variance is

$$s_{n,\lambda}^2 = \mathbb{V}\big(\lambda^\top W_n\big) = \mathbb{V}\big(\lambda^\top \xi_{n,1}\big) = \lambda^\top\big(V \circ \varrho_n\big)\lambda \longrightarrow \sigma_\lambda^2.$$

If $\sigma_\lambda^2 = 0$, then, from Chebyshev's inequality, for every $\varepsilon > 0$, $\mathbb{P}\big(\big|\lambda^\top W_n\big| > \varepsilon\big) \le s_{n,\lambda}^2/\varepsilon^2 \to 0$, so that $\lambda^\top W_n \xrightarrow{\mathbb{P}} 0$ and (S22) holds, the limit law being degenerate at zero. Assume in the sequel that $\sigma_\lambda^2 > 0$.

Step 3: Lindeberg–Feller. For each n, the summands in $\lambda^\top W_n = n^{-1/2}\sum_{j=1}^n \lambda^\top \xi_{n,j}$ are i.i.d. with zero mean and variance $s_{n,\lambda}^2 \to \sigma_\lambda^2 > 0$. Hence, from the Lindeberg–Feller central limit theorem for triangular arrays, (S22) follows if

$$L_n(\varepsilon) = \frac{1}{n}\sum_{j=1}^n \mathbb{E}\Big\{\big(\lambda^\top \xi_{n,j}\big)^2\, I\big(\big|\lambda^\top \xi_{n,j}\big| > \varepsilon\sqrt{n}\big)\Big\} = \mathbb{E}\Big\{\big(\lambda^\top \xi_{n,1}\big)^2\, I\big(\big|\lambda^\top \xi_{n,1}\big| > \varepsilon\sqrt{n}\big)\Big\} \longrightarrow 0, \quad \forall\, \varepsilon > 0, \tag{S23}$$

where $I(\cdot)$ denotes the indicator function. The second equality holds because $\xi_{n,1}, \ldots, \xi_{n,n}$ are identically distributed. Because $\mathbb{E}\{(\lambda^\top \xi_{n,j})^2\} = \sigma_\lambda^2 < \infty$, the convergence to 0 in expression (S23) is obtained by applying the dominated convergence theorem. Hence (S22) holds for every $\lambda \in \mathbb{R}^q$, and the Cramér–Wold device yields $W_n \xrightarrow{\mathcal{L}} \mathcal{N}_q\big(0, V \circ \varrho\big)$, which completes the proof. □

Proof of Corollary 2. Note that $\widehat{\mathcal{D}}_n\big(\widehat{\eta}_{AC} - \eta\big) = \widehat{\mathcal{D}}_n\mathcal{D}_n^{-1}\mathcal{D}_n\big(\widehat{\eta}_{AC} - \eta\big)$. By Theorem 1, it suffices to show that $\widehat{\mathcal{D}}_n\mathcal{D}_n^{-1} \xrightarrow{\mathbb{P}} I_q$, where I_q denotes the $q \times q$ identity matrix, and then apply Slutsky's theorem.

Now,

$$\widehat{\mathcal{D}}_n\mathcal{D}_n^{-1} = \operatorname{diag}\left(\sqrt{\frac{N^{(1)}}{n\pi_n^{(1)}}}, \ldots, \sqrt{\frac{N^{(q)}}{n\pi_n^{(q)}}}\right).$$

Fix $1 \leq r \leq q$. From (S1), we have

$$\mathbb{E}\left(\frac{N^{(r)}}{n\pi_n^{(r)}}\right) = 1 \quad \text{and} \quad \mathbb{V}\left(\frac{N^{(r)}}{n\pi_n^{(r)}}\right) = \frac{1-\pi_n^{(r)}}{n\pi_n^{(r)}} \longrightarrow 0, \tag{S24}$$

where the convergence follows from Assumption A4. Hence, by Chebyshev's inequality,

$$N^{(r)}/\left(n\pi_n^{(r)}\right) \xrightarrow{\mathbb{P}} 1. \tag{S25}$$

Since q is fixed, we conclude that $\widehat{\mathcal{D}}_n\mathcal{D}_n^{-1} \xrightarrow{\mathbb{P}} I_q$. □

Proof of Corollary 3. Since $N^{(r)}/N = \{N^{(r)}/(n\pi_n^{(r)})\}\{\pi_n^{(r)}/\sum_{s=1}^q \pi_n^{(s)}\}$, (S25) and Assumption A6 give $N^{(r)}/N \xrightarrow{\mathbb{P}} \kappa_r$, $1 \leq r \leq q$, and hence $N^{1/2}\widehat{\mathcal{D}}_n^{-1} \xrightarrow{\mathbb{P}} K$ and $\widehat{N}/N = \sum_{r=1}^q N^{(r)}/N \xrightarrow{\mathbb{P}} 1$. Writing $\sqrt{N}(\widehat{\eta}_{AC} - \eta) = N^{1/2}\widehat{\mathcal{D}}_n^{-1}\,\widehat{\mathcal{D}}_n(\widehat{\eta}_{AC} - \eta)$, part 1 follows from Corollary 2 and Slutsky's theorem, since $K(V \circ \varrho)K = V \circ (K\varrho K) = V \circ M$, K being diagonal. Part 2 follows from part 1 and $\widehat{N}/N \xrightarrow{\mathbb{P}} 1$. □

Proof of Corollary 5. The result follows from Corollary 3 and the multivariate Delta Method (see e.g. Theorem 3.3.A of Serfling (1980)). □

Lemma 5. *Suppose that Assumptions A1-A5 hold. Let $D_r \subseteq \{1, \ldots, n\}$ denote the set of data indices used for the calculation of $\widehat{\eta}_{AC}^{(r)}$. For $j \in D_r$, let $\widehat{\eta}_{AC,-j}^{(r)}$ denote the jackknife version leaving out observation X_j. Then $\max_{j \in D_r} \left|\widehat{\eta}_{AC,-j}^{(r)} - \widehat{\eta}_{AC}^{(r)}\right| = O_{\mathbb{P}}\left((\pi_n^{(r)}n)^{-1/2}\right)$, $1 \leq r \leq q$.*

Proof. In order to simplify notation we suppress the super index $^{(r)}$ in $\widehat{\eta}_{AC}^{(r)}$, $N^{(r)}$, $h^{(r)}$ and $\pi_n^{(r)}$, and the subindex $_r$ in D_r and m_r. Moreover, w.l.o.g. it will be assumed that $\eta = 0$ and $D = \{1, 2, \ldots, N\}$. Notice that here N is a short for $N^{(r)}$, having a different meaning as in the statement of Corollary 3.

For each $1 \leq j \leq N$ we have that $\widehat{\eta}_{AC,-j} = \binom{N}{m}\binom{N-1}{m}^{-1}\widehat{\eta}_{AC} - mY_j/(N-m) = N\widehat{\eta}/(N-M)_{AC} - mY_j/(N-m)$, where

$$Y_j = \binom{N-1}{m-1}^{-1} \sum_{1 \leq i_1 < \cdots < i_{m-1} \leq N,\ i_1, \ldots, i_{m-1} \neq j} h(X_j, X_{i1}, \ldots, X_{i_{m-1}}),$$

and therefore,

$$\max_{j \in D} \left|\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\right| \leq \max_{j \in D} \frac{m}{N-m}|Y_j| + \frac{N}{N-m}\left|\widehat{\eta}_{AC}\right|. \tag{S26}$$

From (S24), it follows that $N/(N-m) \xrightarrow{\mathbb{P}} 1$. From Theorem 1, we have that $\widehat{\eta}_{AC} = O_{\mathbb{P}}\left((n\pi_n)^{-1/2}\right)$. Therefore,

$$N\left|\widehat{\eta}_{AC}\right|/(N-m) = O_{\mathbb{P}}\left((n\pi_n)^{-1/2}\right). \tag{S27}$$

On the other hand, recalling that we are working on the set $\{N \geq m\}$, and that, given that $N = \ell$, for some $m \leq \ell \leq n$, the random variables $Y_1, \ldots, Y_\ell$ are exchangeable, we have that

$$\begin{aligned}\mathbb{P}\left(\max_{j\in D}\sqrt{n\pi_n}\frac{m}{N-m}\left|Y_j\right| > \varepsilon\right) &= \sum_{\ell=m}^{n}\mathbb{P}\left(\max_{j\in D}\sqrt{n\pi_n}\frac{m}{N-m}\left|Y_j\right| > \varepsilon \,|\, N=\ell\right)\mathbb{P}(N=\ell)\\ &\leq \sum_{\ell=m}^{n}\sum_{j\in D}\mathbb{P}\left(\sqrt{n\pi_n}\frac{m}{N-m}\left|Y_j\right| > \varepsilon \,|\, N=\ell\right)\mathbb{P}(N=\ell) = R_1(n_0) + R_2(n_0),\end{aligned} \tag{S28}$$

where

$$\begin{aligned}R_1(n_0) &= \sum_{\ell=m}^{n_0}\ell\mathbb{P}\left(\sqrt{n\pi_n}\frac{m}{N-m}\left|Y_1\right| > \varepsilon \,|\, N=\ell\right)\mathbb{P}(N=\ell),\\ R_2(n_0) &= \sum_{\ell=n_0+1}^{n}\ell\mathbb{P}\left(\sqrt{n\pi_n}\frac{m}{N-m}\left|Y_1\right| > \varepsilon \,|\, N=\ell\right)\mathbb{P}(N=\ell).\end{aligned}$$

Recall (S1) and take $n_0 = \lfloor n\pi_n/2 \rfloor$, the largest integer less than or equal to $n\pi_n/2$. For this choice of n_0, we have that

$$0 \leq R_1(n_0) \leq n\pi_n\mathbb{P}(N \leq n\pi_n/2) \leq n\pi_n\exp(-n\pi_n/8) \to 0, \tag{S29}$$

where the second inequality follows from the Chernoff bound for the binomial distribution.

Notice that, given that $N = \ell$, Y_j can be seen as a U-statistic conditioned on X_j with kernel $g(x_1, \ldots, x_{m-1}) = h(X_j, x_1, \ldots, x_{m-1})$. Now, applying the total variance formula we get that

$$\mathbb{V}(Y_j|N=\ell) = \mathbb{V}\{\mathbb{E}(Y_j|X_j, N=\ell)\} + \mathbb{E}\{\mathbb{V}(Y_j|X_j, N=\ell)\}. \tag{S30}$$

For the first term we have that $\mathbb{V}\{\mathbb{E}(Y_j|X_j, N=\ell)\} := \zeta_1$. For the second, applying (S58), we have that

$$\mathbb{E}\{\mathbb{V}(Y_j|X_j, N=\ell)\} = \binom{\ell-1}{m-1}^{-1}\sum_{c=1}^{m-1}\binom{m-1}{c}\binom{\ell-m}{m-1-c}\mathbb{E}[\mathbb{V}\{h_{c+1}(X_j, X_1, \ldots, X_c)|X_j\}]. \tag{S31}$$

Now, applying the total variance law, it follows that

$$\zeta_{c+1} := \mathbb{V}\{h_{c+1}(X_j, X_1, \ldots, X_c)\} = \mathbb{E}[\mathbb{V}\{h_{c+1}(X_j, X_1, \ldots, X_c) \mid X_j\}] + \mathbb{V}[\mathbb{E}\{h_{c+1}(X_j, X_1, \ldots, X_c) \mid X_j\}] \geq \mathbb{E}[\mathbb{V}\{h_{c+1}(X_j, X_1, \ldots, X_c) \mid X_j\}].$$

With this, using (S30) and (S31) and that $\zeta_1 \leq \zeta_m = \mathbb{E}\{|h(X_1, \ldots, X_m)|^2\} < \infty$ we get that, for large ℓ, $\mathbb{V}(Y_j|N = \ell) = \zeta_1 + O(1/\ell) \leq 2\zeta_1$. Taking this fact into account, applying Markov's inequality and that $\ell/(\ell - m) \leq 2$, we get

$$0 \leq R_2(n_0) \leq \frac{2\zeta_1 n\pi_n}{\varepsilon^2} \sum_{\ell=n_0+1}^{n} \ell \left(\frac{m}{\ell - m}\right)^2 \mathbb{P}(N = \ell) \leq \frac{8\zeta_1 m^2 n\pi_n}{\varepsilon^2} \sum_{\ell=n_0+1}^{n} \frac{1}{\ell}\mathbb{P}(N = \ell) \leq \frac{16\zeta_1 m^2}{\varepsilon^2} \leq C, \tag{S32}$$

where the last inequality comes from

$$\sum_{\ell=n_0+1}^{n} \frac{1}{\ell}\mathbb{P}(N = \ell) \leq \frac{1}{n_0 + 1} \sum_{\ell=n_0+1}^{n} \mathbb{P}(N = \ell) \leq \frac{1}{n_0 + 1} < \frac{1}{n_0} \leq \frac{2}{n\pi}.$$

From (S28), (S29) and (S32), we get that $\max_{j \in D} m|Y_j|/(N - m) = O_{\mathbb{P}}\left((n\pi_n)^{-1/2}\right)$. Combining this result with (S26) and (S27) yields the desired result. □

Proof of Proposition 4. Each component of f depends on a subset of components of η. For $1 \leq r \leq d$, let $L_r \subseteq \{1, \ldots, q\}$ be the set of indices such that $f^{(r)}(\eta)$ depends on η only through the subvector $\eta^{L_r} = (\eta^{(l)})_{l \in L_r}^\top$. Let $L_{rs} = L_r \cup L_s$.

In order to simplify notation, along the proof, $\widehat{\eta}_{AC,-j}^{L_{rs}}(D_{rs}^f) = \widehat{\eta}_{AC,-j}$, $\widehat{\eta}_{AC}^{L_{rs}}(D_{rs}^f) = \widehat{\eta}_{AC}(D_{rs}^f)$, $\nabla f^{(v),L_{rs}} = \nabla f^{(v)}$. Let $\pi_{rs,n}^f$ such that $n_{rs}^f \sim \text{Bin}(n, \pi_{rs,n}^f)$. Recall that we are assuming $\mathbb{E}(n_{rs}^f) = n\pi_{rs,n}^f \to \infty$. Since f has continuous first-order partial derivatives, for each $j \in D_{rs}^f$, we have that

$$f^{(v)}\left(\widehat{\eta}_{AC,-j}\right) = f^{(v)}\left(\widehat{\eta}_{AC}\right) + \left(\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\right)^\top \nabla f^{(v)}(\tau_{j,v}), \quad v = r, s,$$

where $\tau_{,v}$ lies between $\widehat{\eta}_{AC,-j}$ and $\widehat{\eta}_{AC}$. By definition,

$$\widehat{\sigma}_{rs} = (n_{rs}^f - 1) \sum_{j \in D_{rs}^f} \left\{ f^{(r)}\left(\widehat{\eta}_{AC,-j}\right) - \frac{1}{n_{rs}^f} \sum_{t \in D_{rs}} f^{(r)}\left(\widehat{\eta}_{AC,-t}\right) \right\} \left\{ f^{(s)}\left(\widehat{\eta}_{AC,-j}\right) - \frac{1}{n_{rs}^f} \sum_{t \in D_{rs}} f^{(s)}\left(\widehat{\eta}_{AC,-t}\right) \right\}.$$

Noting that $\sum_{j \in D_{rs}^f} (\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}) = 0$, we can write,

$$\widehat{\sigma}_{rs} = (n_{rs}^f - 1) \sum_{j \in D_{rs}^f} \Big[\left(\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\right)^\top \nabla f^{(r)}(\eta) + \left(\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\right)^\top \nabla \left\{ f^{(r)}(\tau_{j,r}) - f^{(r)}(\eta) \right\} - \frac{1}{n_{rs}^f} \sum_{w \in D_{rs}^f} \left(\widehat{\eta}_{AC,-w} - \widehat{\eta}_{AC}\right)^\top \nabla \left\{ f^{(r)}(\tau_{w,r}) - f^{(r)}(\eta) \right\} \Big] \Big[\left(\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\right)^\top \nabla f^{(s)}(\eta)$$

$$+ (\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC})^\top \nabla \left\{ f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta) \right\} - \frac{1}{n_{rs}^f} \sum_{w \in D_{rs}^f} (\widehat{\eta}_{AC,-w} - \widehat{\eta}_{AC})^\top \nabla \left\{ f^{(s)}(\tau_{w,s}) - f^{(b)}(\eta) \right\} \Bigg] .$$

The previous expression contains terms of five different forms:

$$\begin{aligned}
T_1 =& (n_{rs}^f - 1) \sum_{j \in D_{rs}^f} (\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC})^\top \nabla f^{(r)}(\eta) \ (\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC})^\top \nabla f^{(s)}(\eta), \\
T_2 =& (n_{rs}^f - 1) \sum_{j \in D_{rs}^f} (\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC})^\top \nabla f^{(r)}(\eta) \ (\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC})^\top \nabla \left\{ f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta) \right\}, \\
T_3 =& (n_{rs}^f - 1) \frac{1}{n_{rs}^f} \sum_{j \in D_{rs}^f} \sum_{w \in D_{rs}^f} (\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC})^\top \nabla f^{(r)}(\eta) \ (\widehat{\eta}_{AC,-w} - \widehat{\eta}_{AC})^\top \nabla \left\{ f^{(s)}(\tau_{w,s}) - f^{(s)}(\eta) \right\}, \\
T_4 =& (n_{rs}^f - 1) \sum_{j \in D_{rs}^f} (\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC})^\top \nabla \left\{ f^{(r)}(\tau_{j,r}) - f^{(r)}(\eta) \right\} \ (\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC})^\top \nabla \left\{ f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta) \right\}, \\
T_5 =& (n_{rs}^f - 1) \frac{1}{n_{rs}^f} \sum_{j \in D_{rs}^f} \sum_{w \in D_{rs}^f} (\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC})^\top \nabla \left\{ f^{(r)}(\tau_{j,r}) - f^{(r)}(\eta) \right\} \\
& \times (\widehat{\eta}_{AC,-w} - \widehat{\eta}_{AC})^\top \nabla \left\{ f^{(s)}(\tau_{w,s}) - f^{(s)}(\eta) \right\}.
\end{aligned}$$

The remainder of the proof will show that T_1 converges in probability to σ_{rs}, while $T_2 - T_5$ converge to zero in probability. We begin with T_1. Let $M = \max\{m_r, m_s\}$ and $\varepsilon > 0$. We have that

$$\mathbb{P}(|T_1 - \sigma_{rz}| > \varepsilon) = \sum_{\ell=M}^{n} \mathbb{P}\left(|T_1 - \sigma_{rz}| > \varepsilon \,\middle|\, |D_{rs}^f| = \ell \right) \mathbb{P}(|D_{rs}^f| = \ell) = S_1(n_0) + S_2(n_0), \tag{S33}$$

for any $M \leq n_0 \leq n$, where

$$\begin{aligned}
S_1(n_0) &= \sum_{\ell=M}^{n_0} \mathbb{P}\left(|T_1 - \sigma_{rz}| > \varepsilon \,\middle|\, |D_{rs}^f| = \ell \right) \mathbb{P}(|D_{rs}^f| = \ell), \\
S_2(n_0) &= \sum_{\ell=n_0+1}^{n} \mathbb{P}\left(|T_1 - \sigma_{rz}| > \varepsilon \,\middle|\, |D_{rs}^f| = \ell \right) \mathbb{P}(|D_{rs}^f| = \ell).
\end{aligned}$$

The proof of Lemma 1 in Romero-Madroñal et al. (2026) shows that, for the case where there is no missing data, $T_1 - \sigma_{rs} \xrightarrow{\mathbb{P}} 0$, which implies that $\forall \delta > 0$, $\exists n_1 = n_1(\varepsilon, \delta)$, such that $\mathbb{P}(|T_1 - \sigma_{rs}| > \varepsilon \,|\, |D_{rs}^f| = \ell) < \delta$, $\forall \ell > n_1$, and thus

$$S_2(n_1) \leq \delta. \tag{S34}$$

As $\mathbb{E}(n_{rs}^f) \to \infty$, for large enough n, $n_1 \leq E(|D_{rs}^f|)/2$, and from the Chernoff bound for

the binomial distribution,

$$0 \leq S_1(n_1) \leq \mathbb{P}\{|D_{rs}^f| \leq 0.5E(|D_{rs}^f|)\} \leq \exp(|D_{rs}^f|/2) \to 0, \tag{S35}$$

Since δ is arbitrary, the results in (S33)–(S35) give that $T_1 \xrightarrow{\mathbb{P}} \sigma_{rs}$.

Applying the Cauchy–Schwarz inequality, the absolute value of T_2 can be bounded by

$$(n_{rs}^f - 1) \sum_{j \in D_{rs}^f} \|\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\|^2 \|\nabla f^{(r)}(\eta)\| \left\|\nabla \left\{f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta)\right\}\right\|. \tag{S36}$$

By assumption $\|\nabla f^{(r)}(\eta)\|_2$ is bounded. That $(n_{rs}^f - 1)\sum_{j \in D_{rs}^f} \|\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\|^2 = O_{\mathbb{P}}(1)$ comes from the previous case taking f the identity function. Finally, from Lemma 5 and Theorem 1, it follows that

$$\max_{j \in D_{rs}^f} \|\tau_{j,v} - \eta\| \leq \max_{j \in D_{rs}^f} \|\widehat{\eta}_{AC} - \widehat{\eta}_{AC,-j}\| + \|\widehat{\eta}_{AC} - \eta\| = O_{\mathbb{P}}\left(\{\mathbb{E}(n_{rs}^f)\}^{-1/2}\right) + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1), v = r, s.$$

With this, given the fact that $f^{(s)}$ has continuous first-order partial derivatives near η, we get that

$$\max_{j \in D_{rs}^f} \left\|\nabla \left\{f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta)\right\}\right\| = o_{\mathbb{P}}(1).$$

Therefore, $T_2 \xrightarrow{\mathbb{P}} 0$. Now, we study T_3. Applying the Cauchy–Schwarz inequality we have that its absolute value can be bounded by

$$\begin{aligned}(n_{rs}^f - 1)(n_{rs}^f)^{-1} &\left(\sum_{j \in D_{rs}^f} \|\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\|\right)^2 \left\|\nabla f^{(r)}(\eta)\right\| \max_{j \in D_{rs}^f} \left\|\nabla \left\{f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta)\right\}\right\| \\ &\leq (n_{rs}^f - 1) \sum_{j \in D_{rs}^f} \|\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\|^2 \left\|\nabla f^{(r)}(\eta)\right\| \max_{j \in D_{rs}^f} \left\|\nabla \left\{f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta)\right\}\right\|\end{aligned}$$

By applying the same arguments as in the previous case, we have that $T_3 \xrightarrow{\mathbb{P}} 0$. Now, applying the Cauchy–Schwarz inequality and taking into account that

$$\begin{aligned}&\max_{j \in D_{rs}^f} \left\|\nabla \left\{f^{(r)}(\tau_{j,r}) - f^{(r)}(\eta)\right\}\right\| \left\|\nabla \left\{f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta)\right\}\right\| \leq \\ &\max_{j \in D_{rs}^f} \left\|\nabla \left\{f^{(r)}(\tau_{j,r}) - f^{(r)}(\eta)\right\}\right\| \max_{j \in D_{rs}^f} \left\|\nabla \left\{f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta)\right\}\right\|,\end{aligned}$$

$|T_4|$ can be bounded by

$$(n_{rs}^f - 1) \left(\sum_{j \in D_{rs}^f} \|\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\|^2\right) \max_{j \in D_{rs}^f} \left\|\nabla \left\{f^{(r)}(\tau_{j,r}) - f^{(r)}(\eta)\right\}\right\| \max_{j \in D_{rs}^f} \left\|\nabla \left\{f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta)\right\}\right\|.$$

The same arguments as in the previous cases apply here and thus $T_4 \xrightarrow{\mathbb{P}} 0$. Finally, applying the Cauchy–Schwarz inequality, $|T_5|$ can be bounded by

$$(n_{rs}^f - 1)\left(\sum_{j\in D_{rs}} \|\widehat{\eta}_{AC,-j} - \widehat{\eta}_{AC}\|^2\right) \max_{j\in D_{rs}^f} \left\|\nabla\left\{f^{(r)}(\tau_{j,r}) - f^{(r)}(\eta)\right\}\right\| \max_{j\in D_{rs}^f} \left\|\nabla\left\{f^{(s)}(\tau_{j,s}) - f^{(s)}(\eta)\right\}\right\|.$$

The same arguments apply here and then $T_5 \xrightarrow{\mathbb{P}} 0$. □

Proof of Proposition 5. First, we decompose

$$\widehat{N}^{1/2}\, Cf\big(\widehat{\eta}_{AC}\big) = \widehat{N}^{1/2}\, C\Big\{f\big(\widehat{\eta}_{AC}\big) - f(\eta)\Big\} + \Big(\frac{\widehat{N}}{N}\Big)^{1/2} N^{1/2} Cf(\eta).$$

From Corollary 5, the first term is $O_{\mathbb{P}}(1)$. A similar argument to that of the proof of Corollary 2, gives that $\widehat{N}/N \xrightarrow{\mathbb{P}} 1$. So, if $N^{1/2}Cf(\eta) \to b \in \mathbb{R}^c$, the second term converges in probability to b. Hence, from Slutsky's theorem, $\widehat{N}^{1/2}\, Cf\big(\widehat{\eta}_{AC}\big) \xrightarrow{\mathcal{L}} Y + b$, where $Y \sim \mathcal{N}_c(0, \Sigma_f)$. If $\|N^{1/2}Cf(\eta)\| \to \infty$, then $\Big(\widehat{N}/N\Big)^{1/2} N^{1/2}Cf(\eta) \xrightarrow{\mathbb{P}} \infty$. Therefore,

$$\big\|\widehat{N}^{1/2}Cf(\widehat{\eta}_{AC})\big\| \geq \Big\|\Big(\frac{\widehat{N}}{N}\Big)^{1/2} N^{1/2}Cf(\eta)\Big\| - \Big\|\widehat{N}^{1/2}C\{f(\widehat{\eta}_{AC}) - f(\eta)\}\Big\| \xrightarrow{\mathbb{P}} \infty,$$

because the first term diverges while the second one is $O_{\mathbb{P}}(1)$. On the other hand, $\widehat{\eta}_{AC} \xrightarrow{\mathbb{P}} \eta$ and the first-order partial derivatives of f are continuous in a neighbourhood of η, so that $\nabla f(\widehat{\eta}_{AC}) \xrightarrow{\mathbb{P}} \nabla f(\eta)$. As for the matrix $\widehat{N}\,\widehat{\Omega} = \Big(\widehat{N}\,\widehat{\Omega}_{rs}\Big)$, note that

$$\widehat{N}\,\widehat{\Omega}_{rs} = \widehat{V}_{rs}\frac{\widehat{N}}{n}\frac{N^{(r,s)}/n}{\big(N^{(r)}/n\big)\big(N^{(s)}/n\big)}.$$

From (S25), $N^{(r)}/(n\pi^{(r)}) \xrightarrow{\mathbb{P}} 1$, and analogously $N^{(r,s)}/(n\pi^{(r,s)}) \xrightarrow{\mathbb{P}} 1$, $1 \leq r, s \leq q$, while $\widehat{V}_{rs} \xrightarrow{\mathbb{P}} V_{rs}$ from Corollary 6. Hence

$$\widehat{N}\,\widehat{\Omega}_{rs} \xrightarrow{\mathbb{P}} V_{rs}\Big(\sum_u \pi^{(u)}\Big)\frac{\pi^{(r,s)}}{\pi^{(r)}\pi^{(s)}} = V_{rs}M_{rs},$$

the last equality coming from the definition of M (see Corollary 3). Therefore, $\widehat{N}\,\widehat{\Omega} \xrightarrow{\mathbb{P}} V \circ M$, and consequently $\widehat{N}C\widehat{\Omega}_f C^\top \xrightarrow{\mathbb{P}} \Sigma_f$. Since Σ_f is nonsingular and the set of nonsingular matrices is open, with inversion continuous on it, $\widehat{N}C\widehat{\Omega}_f C^\top$ is nonsingular with probability tending to one, its pseudoinverse coincides there with the ordinary inverse, and $(\widehat{N}C\widehat{\Omega}_f C^\top)^- \xrightarrow{\mathbb{P}} \Sigma_f^{-1}$. Finally, since $(cA)^- = c^{-1}A^-$ for every $c > 0$, the statistic

can be written as

$$T_n = \Big\{\widehat{N}^{1/2} C f(\widehat{\eta}_{AC})\Big\}^\top (\widehat{N} C \widehat{\Omega}_f C^\top)^- \Big\{\widehat{N}^{1/2} C f(\widehat{\eta}_{AC})\Big\}.$$

In the first case, the Continuous Mapping Theorem together with Theorem 4.4.3 of Graybill (1976) yields the result. In the second case, the result is obtained since $(\widehat{N} C \widehat{\Omega}_f C^\top)^- \xrightarrow{\mathbb{P}} \Sigma_f^{-1}$ and Σ_f^{-1} is positive definite. □

Proof of Proposition 6. Write $\widehat{N}_i = \sum_s N_i^{(s)}$ and $\widehat{N} = \sum_i \widehat{N}_i$. As in the proof of Proposition 5, $\widehat{N}_i / N_i \xrightarrow{\mathbb{P}} 1$, and therefore $\widehat{N}/N \xrightarrow{\mathbb{P}} 1$ and $\widehat{N}/\widehat{N}_i \xrightarrow{\mathbb{P}} 1/\lambda_i$, by Assumption A8. Decompose

$$\widehat{N}^{1/2}\, C f(\widehat{\eta}) = \widehat{N}^{1/2}\, C\big\{f(\widehat{\eta}) - f(\eta)\big\} + \Big(\frac{\widehat{N}}{N}\Big)^{1/2} N^{1/2} C f(\eta).$$

For the first term, within each sample Corollary 5 gives $\widehat{N}_i^{1/2}\big\{f(\widehat{\eta}_{AC,i}) - f(\eta_i)\big\} \xrightarrow{\mathcal{L}} \mathcal{N}_d\big(0, \Sigma_{f,i}\big)$, so that $\widehat{N}^{1/2}\big\{f(\widehat{\eta}_{AC,i}) - f(\eta_i)\big\} = \big(\widehat{N}/\widehat{N}_i\big)^{1/2} \widehat{N}_i^{1/2}\big\{f(\widehat{\eta}_{AC,i}) - f(\eta_i)\big\} \xrightarrow{\mathcal{L}} \mathcal{N}_d\big(0, \Sigma_{f,i}/\lambda_i\big)$; as the k samples are independent, the joint convergence $\widehat{N}^{1/2}\, C\big\{f(\widehat{\eta}) - f(\eta)\big\} \xrightarrow{\mathcal{L}} Y \sim \mathcal{N}_{kd}\big(0, \Sigma_f\big)$ follows. As for the matrix $C\,\widehat{\Omega}_f\, C^\top$, applying within each sample the argument used for the proof of Proposition 5 yields $\widehat{N}_i\, \widehat{\Omega}_{f,i} \xrightarrow{\mathbb{P}} \Sigma_{f,i}$, whence $\widehat{N}\, \widehat{\Omega}_{f,i} = \big(\widehat{N}/\widehat{N}_i\big)\widehat{N}_i \widehat{\Omega}_{f,i} \xrightarrow{\mathbb{P}} \Sigma_{f,i}/\lambda_i$ and, the blocks being arranged diagonally, $\widehat{N}\, C \widehat{\Omega}_f C^\top \xrightarrow{\mathbb{P}} \Sigma_f$. Now, the same arguments used in the proof of Proposition 5 give the desired results. □

Proof of Theorem 2. First, suppose $\kappa_{full} = 0$. By definition (20), $\delta_{CC} = 0$. Furthermore, $\delta_{AC} = b^\top \Sigma_f^{-1} b \geq 0$, with equality if and only if $b = 0$, since the matrix Σ_f^{-1} is positive definite. This proves part 1.

Assume now that $\kappa_{full} > 0$, and define $B = \frac{1}{\kappa_{full}} C\, \nabla f(\eta) V \nabla f(\eta)^\top C^\top$, so that $\delta_{CC} = b^\top B^{-1} b$. A direct calculation yields

$$\big(V/\kappa_{full} - V \circ M\big)_{rs} = V_{rs}\left(\frac{1}{\kappa_{full}} - M_{rs}\right) = \big(V \circ Q\big)_{rs}, \quad 1 \leq r, s \leq q.$$

Consequently, $B - \Sigma_f = C \nabla f(\eta) \big(V \circ Q\big) \nabla f(\eta)^\top C^\top$.

By definition, $\delta_{AC} \geq \delta_{CC}$ for every $b \in \mathbb{R}^c$ if and only if the symmetric matrix $\Sigma_f^{-1} - B^{-1}$ is positive semidefinite, which we denote as $\Sigma_f^{-1} \succeq B^{-1}$. For positive definite matrices A_1 and A_2, it is a standard result that $A_1 \succeq A_2$ if and only if $A_2^{-1} \succeq A_1^{-1}$ (see, e.g., Corollary 7.7.4 (a) in Horn and Johnson (2012)). Applying this equivalence, $\Sigma_f^{-1} \succeq B^{-1}$ holds if and only if $B \succeq \Sigma_f$, which means that $B - \Sigma_f = C \nabla f(\eta) \big(V \circ Q\big) \nabla f(\eta)^\top C^\top$ is positive semidefinite. □

Proof of Corollary 7. If $\kappa_{full} = 0$, both claims follow trivially from part 1 of Theorem 2,

so we assume $\kappa_{full} > 0$. From (5), we have $V_{rs} = m_r m_s \Gamma_{rs} = 0$, $r \neq s$. Consequently,

$$V \circ Q = \operatorname{diag}\big(V_{11}Q_{11}, \ldots, V_{qq}Q_{qq}\big).$$

Since $M_{rr} = 1/\kappa_r$ and $\kappa_{full} \leq \min_{1\leq r\leq q} \kappa_r$, it follows that $Q_{rr} = 1/\kappa_{full} - 1/\kappa_r \geq 0$ for $1 \leq r \leq q$. Thus, $V \circ Q$ is positive semidefinite. Consequently, the matrix $C\nabla f(\eta)\big(V \circ Q\big)\nabla f(\eta)^\top C^\top$ is also positive semidefinite, and part 2 of Theorem 2 yields the result. □

Proof of Corollary 8. The diagonal entries of $V \circ Q$ are $v\,Q_{rr}$ and the off-diagonal ones are $v\rho\,Q_{rs}$, $r \neq s$. Thus,

$$C\nabla f\big(\eta\big)\big(V \circ Q\big)\nabla f\big(\eta\big)^\top C^\top = v\big\{\big(1-\rho\big)\Psi_D + \rho\,\Psi_Q\big\},$$

where $\Psi_Q = C\nabla f\big(\eta\big)Q\nabla f\big(\eta\big)^\top C^\top$ and Ψ_D is defined in the statement of the result. Since $v > 0$, Part 2 of Theorem 2 implies that $\rho \in \mathcal{R}$ if and only if $\big(1-\rho\big)\Psi_D + \rho\,\Psi_Q \succeq 0$. The matrix mapping $\rho \mapsto (1-\rho)\Psi_D + \rho\,\Psi_Q$ is affine (and hence continuous), and the cone of positive semidefinite matrices is closed and convex. Consequently, the set of values $\rho \in \mathbb{R}$ for which this matrix is positive semidefinite is a closed interval in $\mathbb{R}$. Intersecting this set with the domain ι_q establishes that $\mathcal{R}$ is a relatively closed interval in I. From Corollary 7, we have $0 \in \mathcal{R}$. Let $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote the minimum and maximum eigenvalues of a real symmetric matrix, respectively. If Ψ_D is positive definite, then $\lambda_{\min}(\Psi_D) > 0$. From Weyl's inequality for matrix eigenvalues (e.g., Horn and Johnson, 2012, Theorem 4.3.1), we bound the minimum eigenvalue of the matrix sum from below as follows,

$$\lambda_{\min}\big\{\big(1-\rho\big)\Psi_D + \rho\,\Psi_Q\big\} \geq \lambda_{\min}\big(\big(1-\rho\big)\Psi_D\big) + \lambda_{\min}\big(\rho\,\Psi_Q\big),$$

and then

$$\lambda_{\min}\big\{\big(1-\rho\big)\Psi_D + \rho\,\Psi_Q\big\} \geq \big(1-\rho\big)\lambda_{\min}\big(\Psi_D\big) - \big|\rho\big|\max\big\{\big|\lambda_{\min}(\Psi_Q)\big|,\,\big|\lambda_{\max}(\Psi_Q)\big|\big\}.$$

For sufficiently small $|\rho|$, the positive term $(1-\rho)\lambda_{\min}(\Psi_D)$ strictly dominates the right-hand side of the above inequality, ensuring that the minimum eigenvalue remains strictly positive. Thus, an open neighborhood of 0 is contained in $\mathcal{R}$, proving that 0 is an interior point of $\mathcal{R}$. This proves part 1.

For part 2, assume that $G = C\nabla f(\eta)$ satisfies $G\mathbb{1}_q = 0$, so that $GJ_qG^\top = 0$ and $GVG^\top = v(1-\rho)GG^\top$. Moreover, $V \circ M = v\{(1-\rho)\operatorname{diag}(M) + \rho M\}$, where $\operatorname{diag}(M)$ denotes the diagonal matrix with the same diagonal as M, so that $\Sigma_f = G(V \circ M)G^\top = v(1-\rho)A(\rho)$, with

$$A(\rho) = G\operatorname{diag}(M)G^\top + \frac{\rho}{1-\rho}\,GMG^\top.$$

Hence, from (20), for every $\rho \in \iota_q$,

$$(1-\rho)\,\delta_{CC} = \frac{\kappa_{full}}{v}\, b^\top (GG^\top)^{-1} b, \qquad (1-\rho)\,\delta_{AC} = \frac{1}{v}\, b^\top A(\rho)^{-1} b.$$

The quantity $(1-\rho)\,\delta_{CC}$ does not depend on ρ. $A(\rho) = \Sigma_f/\{v(1-\rho)\}$ is positive definite for every $\rho \in \iota_q$, and, for $\rho_1 < \rho_2$ in ι_q,

$$A(\rho_2) - A(\rho_1) = \Big(\frac{\rho_2}{1-\rho_2} - \frac{\rho_1}{1-\rho_1}\Big) GMG^\top \succeq 0,$$

because $\rho/(1-\rho)$ is increasing and $GMG^\top$ is positive semidefinite. Consequently, $g(\rho) = (1-\rho)(\delta_{AC} - \delta_{CC})$ is non-increasing on ι_q, with $g(0) \geq 0$ by Corollary 7. Since $1-\rho > 0$, $\delta_{AC} - \delta_{CC}$ has the sign of $g(\rho)$, so $\mathcal{R}_b = \{\rho \in \iota_q : g(\rho) \geq 0\}$ is either the whole ι_q or an interval $(-1/\{q-1\}, \rho^*(b)]$, with $\rho^*(b) \geq 0$ because $g(0) \geq 0$.

□

Before proving Theorem 3, we present some previous results.

Proof of Lemma 1. Consider the enlarged system of $2q$ U-statistics with kernels $h^{(r)}$ and $h^{(q+r)} = h^{(r)}$ and index sets A_r and $A_{q+r} = A_{full}$, $1 \leq r \leq q$, whose AC estimator is the stacked vector $\big(\widehat{\eta}_{AC}^\top, \widehat{\eta}_{CC}^\top\big)^\top$. Assumptions A1 and A2 for the enlarged system coincide with the original ones, because the kernels coincide pairwise; Assumption A4 requires in addition $n\pi_{full} \to \infty$, which holds by Assumption A9; and, since $R_j^{full} \leq R_j^{(r)}$, the joint availability probabilities of the enlarged system involving one original and one enlarged component are π_{full}, so that

$$\tilde{\varrho}_n^{(r,q+s)} = \frac{\pi_{full}}{\sqrt{\pi_n^{(r)}\pi_{full}}} = \left\{\frac{\big(\pi_{full}\big)_n}{\pi_n^{(r)}}\right\}^{1/2} = \left[\left\{\big(\pi_{full}\big)_n \Big/ \sum_{u=1}^{q} \pi_n^{(u)}\right\}\left\{\pi_n^{(r)} \Big/ \sum_{u=1}^{q} \pi_n^{(u)}\right\}^{-1}\right]^{1/2} \to \sqrt{\frac{\kappa_{full}}{\kappa_r}},$$

by Assumptions A6 and A9, while $\tilde{\varrho}_n^{(q+r,q+s)} = 1$; hence Assumption A5 holds for the enlarged system. The claim follows from Corollary 2 applied to it, all four $q \times q$ blocks of $\tilde{V}$ being equal to V because the kernels coincide pairwise. □

Lemma 6. *Let $X_1, \ldots, X_N$ be i.i.d. random elements, let U_N a U-statistic of degree $m \geq 1$ with symmetric kernel h such that $\sigma_m^2 = \mathbb{V}\big\{h\big(X_1, \ldots, X_m\big)\big\} < \infty$, let $\theta = \mathbb{E}\big\{h\big(X_1, \ldots, X_m\big)\big\}$, and let $U_{N-1}^{(-j)}$ denote the same statistic computed without the j-th observation, $N > m$. Define $\widehat{\psi}_j = \big(N-1\big)\big\{U_N - U_{N-1}^{(-j)}\big\}$, $1 \leq j \leq N$. Then $\sum_{j=1}^N U_{N-1}^{(-j)} = N\,U_N$, which implies that $\sum_{j=1}^N \widehat{\psi}_j = 0$, and $(1/N)\sum_{j=1}^N \big\{\widehat{\psi}_j - m\,\tilde{h}_1\big(X_j\big)\big\}^2 = O_{\mathbb{P}}\big(N^{-1}\big)$, with $\tilde{h}_1$ as in* (S54).

Proof. For $j = 1, \ldots, N$, let q_j denote the average of $h(X_S)$ over all subsets $S \subseteq \{1, \ldots, N\}$ of size m such that $j \in S$. Similarly to the proof of Lemma 5, we have

that

$$NU_N = mq_j + (N-m)U_{N-1}^{(-j)}. \tag{S37}$$

where

$$q_j = \binom{N-1}{m-1}^{-1} \sum_{\substack{S\subseteq\{1,\ldots,N\} \\ |S|=m,\ j\in S}} h(X_S).$$

By the definition of q_j,

$$\sum_{j=1}^{N} q_j = \frac{1}{\binom{N-1}{m-1}} \sum_{j=1}^{N} \sum_{\substack{|S|=m \\ j\in S}} h(X_S).$$

Each subset S has exactly m elements, so each term $h(X_S)$ is counted exactly m times. Hence,

$$\sum_{j=1}^{N} q_j = \frac{m}{\binom{N-1}{m-1}} \sum_{|S|=m} h(X_S) = \frac{N}{\binom{N}{m}} \sum_{|S|=m} h(X_S) = NU_N.$$

Therefore, $(1/N)\sum_{j=1}^{N} q_j = U_N$. Averaging (S37) over j therefore gives

$$NU_N = mU_N + (N-m)(1/N)\sum_{j=1}^{N} U_{N-1}^{(-j)},$$

and consequently

$$\frac{1}{N}\sum_{j=1}^{N} U_{N-1}^{(-j)} = U_N. \tag{S38}$$

It follows from the definition of $\widehat{\psi}_j$ that $\sum_{j=1}^{N}\widehat{\psi}_j = (N-1)\left(NU_N - \sum_{j=1}^{N} U_{N-1}^{(-j)}\right) = 0$. Next, (S37) implies that $\widehat{\psi}_j = (N-1)\{U_N - U_{N-1}^{(-j)}\} = mq_j - (m-1)U_{N-1}^{(-j)} - U_N$. Since $\tilde{h}_1(x) = h_1(x) - \theta$, we consequently have that

$$\widehat{\psi}_j - m\tilde{h}_1(X_j) = m\{q_j - h_1(X_j)\} - (m-1)\{U_{N-1}^{(-j)} - \theta\} - \{U_N - \theta\}. \tag{S39}$$

From (S57), (S58) and Lemma 2, for a U-statistic of degree m based on $n \geq 2m$ observations, it follows that $\mathbb{V}(U_n) \leq C_m\sigma_m^2/n$, for some constant $C_m < \infty$ depending only on m. Applying this bound to $U_{N-1}^{(-j)}$ and U_N gives

$$\mathbb{E}\{U_{N-1}^{(-j)} - \theta\}^2 = \mathbb{V}(U_{N-1}^{(-j)}) \leq \frac{C_m\sigma_m^2}{N-1}, \tag{S40}$$

and

$$\mathbb{E}\{U_N - \theta\}^2 = \mathbb{V}(U_N) \leq \frac{C_m\sigma_m^2}{N}. \tag{S41}$$

It remains to consider $q_j - h_1(X_j)$. Conditional on X_j, q_j is a U-statistic of degree $m-1$ based on the remaining $N-1$ observations, with kernel $h_j(x_1,\ldots,x_{m-1}) =$

$h(X_j, x_1, \ldots, x_{m-1})$, and conditional expectation $\mathbb{E}(q_j \mid X_j) = h_1(X_j)$. Applying the previous reasoning, there exists a constant $C_{m-1} < \infty$, depending only on m, such that, almost surely, $\mathbb{V}(q_j \mid X_j) \leq \frac{C_{m-1}}{N-1}\mathbb{V}\{h(X_1, \ldots, X_m) \mid X_j\}$. Consequently,

$$\mathbb{E}\{q_j - h_1(X_j)\}^2 = \mathbb{E}\left\{\mathbb{V}(q_j \mid X_j)\right\} \leq \frac{C_{m-1}}{N-1}\mathbb{E}\left[\mathbb{V}\{h(X_1, \ldots, X_m) \mid X_j\}\right] \leq \frac{C_{m-1}\sigma_m^2}{N-1}, \tag{S42}$$

where the last inequality follows from $\mathbb{E}\left[\mathbb{V}\{h(X_1, \ldots, X_m) \mid X_j\}\right] \leq \mathbb{V}\{h(X_1, \ldots, X_m)\} = \sigma_m^2$. For $m = 1$, $q_j = h(X_j)$ and hence $q_j - h_1(X_j) = 0$ identically. Finally, by (S39) and the inequality $(a+b+c)^2 \leq 3(a^2+b^2+c^2)$, together with (S40)–(S42),

$$\mathbb{E}\left[\{\widehat{\psi}_j - m\tilde{h}_1(X_j)\}^2\right] \leq \frac{3m^2 C_{m-1}\sigma_m^2}{N-1} + \frac{3(m-1)^2 C_m \sigma_m^2}{N-1}\frac{3C_m\sigma_m^2}{N}.$$

As the right-hand side of the above inequality is of order N^{-1}, we have that

$$\mathbb{E}\left[\frac{1}{N}\sum_{j=1}^{N}\{\widehat{\psi}_j - m\tilde{h}_1(X_j)\}^2\right] \leq \frac{C'_m\sigma_m^2}{N}$$

for some constant $C'_m < \infty$ depending only on m. Because the quantity inside the expectation is nonnegative, Markov's inequality then yields the result. □

Lemma 7. *For each $\ell \geq \ell_0$, let t_ℓ be a measurable function of ℓ arguments and let $\{Z_i\}_{i\geq 1}$ be i.i.d. random elements such that $t_\ell(Z_1, \ldots, Z_\ell) \xrightarrow{\mathbb{P}} \tau$ as $\ell \to \infty$. Let Y_n and L_n be random variables, L_n integer-valued with $L_n \xrightarrow{\mathbb{P}} \infty$, such that, for every n and every $\ell \geq \ell_0$, the conditional law of Y_n given $\{L_n = \ell\}$ coincides with the law of $t_\ell(Z_1, \ldots, Z_\ell)$. Then $Y_n \mathbb{1}\{L_n \geq \ell_0\} \xrightarrow{\mathbb{P}} \tau$.*

Proof. Let $\varepsilon, \delta > 0$, let $p_\ell = \mathbb{P}\{|t_\ell(Z_1, \ldots, Z_\ell) - \tau| > \varepsilon\}$ and choose $\ell_1 \geq \ell_0$ such that $p_\ell \leq \delta$ for all $\ell \geq \ell_1$. Then $\mathbb{P}\{|Y_n - \tau| > \varepsilon,\, L_n \geq \ell_0\} = \sum_{\ell \geq \ell_0} p_\ell\, \mathbb{P}(L_n = \ell) \leq \delta + \mathbb{P}(L_n < \ell_1) \to \delta$, and the claim follows because δ is arbitrary. □

Lemma 8. *Suppose that Assumptions A1–A7 and A9 hold, with $\kappa_{full} > 0$. Conditionally on $\mathcal{X}_n$, the vector $\left[\{\widehat{\mathcal{D}}_n(\widehat{\eta}^*_{AC} - \widehat{\eta}_{AC})\}^\top, \{N_{full}^{1/2}(\widehat{\eta}^*_{CC} - \widehat{\eta}_{CC})\}^\top\right]^\top$ has mean zero and covariance matrix $\widehat{\Xi}_n$ given blockwise by*

$$\widehat{\Xi}_n^{11}(r,s) = \frac{1}{(N^{(r)}N^{(s)})^{1/2}}\sum_{j \in D_{rs}} \widehat{\psi}_j^{(r)}\widehat{\psi}_j^{(s)}, \quad \widehat{\Xi}_n^{22}(r,s) = \frac{1}{N_{full}}\sum_{j \in D_{full}} \widehat{\varphi}_j^{(r)}\widehat{\varphi}_j^{(s)},$$

$$\widehat{\Xi}_n^{12}(r,s) = \frac{1}{(N^{(r)}N_{full})^{1/2}}\sum_{j \in D_{full}} \widehat{\psi}_j^{(r)}\widehat{\varphi}_j^{(s)}, \quad 1 \leq r, s \leq q,$$

where $\widehat{\psi}_j^{(r)}$ and $\widehat{\varphi}_j^{(s)}$ are defined in (21) and (22), respectively. Moreover, $\widehat{\Xi}_n \xrightarrow{\mathbb{P}} \tilde{V} \circ \tilde{\varrho}$,

with $\tilde{V}$ and $\tilde{\varrho}$ as in Lemma 1, and

$$\max_{j\in D_r}\frac{\widehat{\psi}_j^{(r)2}}{N^{(r)}}\xrightarrow{\mathbb{P}}0,\qquad \max_{j\in D_{full}}\frac{\widehat{\varphi}_j^{(s)2}}{N_{full}}\xrightarrow{\mathbb{P}}0,\qquad 1\le r,s\le q. \tag{S43}$$

Proof. Conditionally on $\mathcal{X}_n$, the bootstrap vector is a linear combination of the independent variables $W_1,\dots,W_n$, which have mean zero and unit variance. Hence its conditional mean is zero. Moreover, using $\mathbb{E}(W_iW_j)=I\{i=j\}$, its conditional covariance matrix is $\widehat{\Xi}_n$ with the stated blockwise representation. We first record the asymptotic behaviour of the relevant sample sizes. By Assumptions A4, A7, and A9, the variables $N^{(r)}$, $N^{(r,s)}$, and N_{full} are binomial with diverging means. Hence, applying Chebyshev's inequality as in (S25), we obtain

$$\frac{N^{(r)}}{n\pi^{(r)}}\xrightarrow{\mathbb{P}}1,\qquad \frac{N^{(r,s)}}{n\pi^{(r,s)}}\xrightarrow{\mathbb{P}}1,\qquad \frac{N_{full}}{n\pi_{full}}\xrightarrow{\mathbb{P}}1.$$

Therefore,

$$\frac{N^{(r,s)}}{(N^{(r)}N^{(s)})^{1/2}}\xrightarrow{\mathbb{P}}\varrho^{(r,s)},\qquad \left(\frac{N_{full}}{N^{(r)}}\right)^{1/2}\xrightarrow{\mathbb{P}}\Delta_{rs}=\left(\frac{\kappa_{full}}{\kappa_r}\right)^{1/2}. \tag{S44}$$

Moreover, since $\kappa_{full}>0$,

$$\frac{N^{(v)}}{N_{full}}\xrightarrow{\mathbb{P}}\frac{\kappa_v}{\kappa_{full}},\qquad v\in\{r,s\},$$

and, because $D_{full}\subseteq D_{rs}$,

$$\frac{N^{(v)}}{N^{(r,s)}}\le\frac{N^{(v)}}{N_{full}}=O_P(1),\qquad v\in\{r,s\}. \tag{S45}$$

We next establish the convergence results needed to control the entries of $\widehat{\Xi}_n$. Under Assumption A3, $\{M_j\}_{j=1}^n$ and $\{X_j\}_{j=1}^n$ are independent. Hence, for any index set $D\subseteq\{1,\dots,n\}$, conditional on $\{M_j\}_{j=1}^n$, the variables $\{X_j : j\in D\}$ are i.i.d. with the distribution of X_1. Consequently, from Lemmas 7 allows results established for deterministic i.i.d. sample sizes to be transferred to the random available sample sizes. Write

$$\widehat{\psi}_j^{(r)}=m_r\tilde{h}_1^{(r)}(X_j^{A_r})+e_j^{(r)},\qquad j\in D_r,$$

and

$$\widehat{\varphi}_j^{(s)}=m_s\tilde{h}_1^{(s)}(X_j^{A_s})+u_j^{(s)},\qquad j\in D_{full}.$$

From Lemmas 6 and 7, we have that

$$\frac{1}{N^{(r)}}\sum_{j\in D_r} e_j^{(r)2} \xrightarrow{\mathbb{P}} 0, \qquad \frac{1}{N_{full}}\sum_{j\in D_{full}} u_j^{(s)2} \xrightarrow{\mathbb{P}} 0. \tag{S46}$$

Furthermore, applying the weak law of large numbers for i.i.d. data with deterministic sample sizes and then Lemma 7, we obtain

$$\frac{1}{N^{(r,s)}}\sum_{j\in D_{rs}} \tilde{h}_1^{(r)}(X_j^{A_r})\tilde{h}_1^{(s)}(X_j^{A_s}) \xrightarrow{\mathbb{P}} \Gamma_{rs}, \tag{S47}$$

where the integrability required for the weak law follows from Assumption A1, (S57) and

$$\mathbb{E}\left|\tilde{h}_1^{(r)}(X_1^{A_r})\tilde{h}_1^{(s)}(X_1^{A_s})\right| \le (\Gamma_{rr}\Gamma_{ss})^{1/2} < \infty.$$

Similarly,

$$\frac{1}{N^{(v)}}\sum_{j\in D_v} \tilde{h}_1^{(v)}(X_j^{A_v})^2 \xrightarrow{\mathbb{P}} \Gamma_{vv} < \infty, \qquad v \in \{r,s\}. \tag{S48}$$

We first show that, if $\{\zeta_j\}_{j\ge 1}$ is an i.i.d. sequence with $\mathbb{E}(\zeta_1^2) < \infty$, then $(1/\ell)\max_{1\le j\le \ell}\zeta_j^2 \xrightarrow{P} 0$. Indeed, for every $\varepsilon > 0$, by the union bound and the identical distribution of the ζ_j's,

$$\mathbb{P}\left(\frac{1}{\ell}\max_{1\le j\le \ell}\zeta_j^2 > \varepsilon\right) \le \ell\mathbb{P}\left(\zeta_1^2 > \varepsilon\ell\right) = \frac{1}{\varepsilon}\mathbb{E}\left[\varepsilon\ell\, I\left\{\zeta_1^2 > \varepsilon\ell\right\}\right] \le \frac{1}{\varepsilon}\mathbb{E}\left[\zeta_1^2\, I\left\{\zeta_1^2 > \varepsilon\ell\right\}\right] \longrightarrow 0,$$

where the last convergence follows from the Dominated Convergence theorem. The above convergence and Lemma 7 gives

$$\max_{j\in D_r}\frac{\tilde{h}_1^{(r)}(X_j^{A_r})^2}{N^{(r)}} \xrightarrow{P} 0, \qquad \max_{j\in D_{full}}\frac{\tilde{h}_1^{(s)}(X_j^{A_s})^2}{N_{full}} \xrightarrow{P} 0. \tag{S49}$$

We now consider the blocks of $\widehat{\Xi}_n$. The (r,s)-entry of the first block has the following expression,

$$\widehat{\Xi}_n^{11}(r,s) = \frac{N^{(r,s)}}{(N^{(r)}N^{(s)})^{1/2}}\frac{1}{N^{(r,s)}}\sum_{j\in D_{rs}}\widehat{\psi}_j^{(r)}\widehat{\psi}_j^{(s)}.$$

Expanding the product gives

$$\frac{1}{N^{(r,s)}}\sum_{j\in D_{rs}}\widehat{\psi}_j^{(r)}\widehat{\psi}_j^{(s)} = m_r m_s\frac{1}{N^{(r,s)}}\sum_{j\in D_{rs}}\tilde{h}_1^{(r)}(X_j^{A_r})\tilde{h}_1^{(s)}(X_j^{A_s}) + \mathrm{Rem}_{rs},$$

where Rem_{rs} contains the three terms involving at least one residual. From Cauchy–

Schwarz's inequality, (S45), (S46) and (S48), we have

$$\left|\frac{1}{N^{(r,s)}}\sum_{j\in D_{rs}}\tilde{h}_1^{(r)}(X_j^{A_r})e_j^{(s)}\right| \le \left\{\frac{N^{(r)}}{N^{(r,s)}}\frac{1}{N^{(r)}}\sum_{j\in D_r}\tilde{h}_1^{(r)}(X_j^{A_r})^2\right\}^{1/2}\left\{\frac{N^{(s)}}{N^{(r,s)}}\frac{1}{N^{(s)}}\sum_{j\in D_s}e_j^{(s)2}\right\}^{1/2} \xrightarrow{\mathbb{P}} 0.$$

The other two residual terms are handled in the same way, using (S45)–(S48). Hence,

$$\frac{1}{N^{(r,s)}}\sum_{j\in D_{rs}}\widehat{\psi}_j^{(r)}\widehat{\psi}_j^{(s)} \xrightarrow{\mathbb{P}} m_r m_s \Gamma_{rs} = V_{rs}.$$

Together with (S44), it follows that $\widehat{\Xi}_n^{11}(r,s) \xrightarrow{\mathbb{P}} \varrho^{(r,s)}V_{rs}$. For the off-diagonal block, the same arguments, now using $D_{full} \subseteq D_r \cap D_s$, yield $(1/N_{full})\sum_{j\in D_{full}}\widehat{\psi}_j^{(r)}\widehat{\varphi}_j^{(s)} \xrightarrow{\mathbb{P}} V_{rs}$. Consequently, from (S44), we get that $\widehat{\Xi}_n^{12}(r,s) \xrightarrow{\mathbb{P}} \Delta_{rs}V_{rs}$. Finally, $(1/N_{full})\sum_{j\in D_{full}}\widehat{\varphi}_j^{(r)}\widehat{\varphi}_j^{(s)} \xrightarrow{\mathbb{P}} V_{rs}$, and therefore $\widehat{\Xi}_n^{22}(r,s) \xrightarrow{\mathbb{P}} V_{rs}$. Since the above holds for every r, s, we conclude that $\widehat{\Xi}_n \xrightarrow{\mathbb{P}} \tilde{V} \circ \tilde{\varrho}$.

It remains to verify (S43). Since $\widehat{\psi}_j^{(r)2} \le 2m_r^2\tilde{h}_1^{(r)}(X_j^{A_r})^2 + 2e_j^{(r)2}$, and $\max_{j\in D_r} e_j^{(r)2} \le \sum_{j\in D_r} e_j^{(r)2}$, we have that

$$\max_{j\in D_r}\frac{\widehat{\psi}_j^{(r)2}}{N^{(r)}} \le 2m_r^2\max_{j\in D_r}\frac{\tilde{h}_1^{(r)}(X_j^{A_r})^2}{N^{(r)}} + \frac{2}{N^{(r)}}\sum_{j\in D_r}e_j^{(r)2} \xrightarrow{\mathbb{P}} 0,$$

from (S46) and (S49). The same argument using $u_j^{(s)}$ instead of $e_j^{(r)}$ gives $\max_{j\in D_{full}}\widehat{\varphi}_j^{(s)2}/N_{full} \xrightarrow{\mathbb{P}} 0$. This proves (S43) and completes the proof. □

Proof of Theorem 3. By Remark 1, the results of Lemma 1 and Lemmas 6–8 apply to both the AC and CC estimators.

Step 1: Joint asymptotic distribution of the bootstrap estimators. By the definition of the bootstrap estimators,

$$\widehat{\mathcal{D}}_n\big(\widehat{\eta}_{AC}^* - \widehat{\eta}_{AC}\big) = \left(\sum_{j=1}^n W_j\frac{\widehat{\psi}_j^{(1)}}{(N^{(1)})^{1/2}}, \ldots, \sum_{j=1}^n W_j\frac{\widehat{\psi}_j^{(q)}}{(N^{(q)})^{1/2}}\right)^\top,$$

where $\widehat{\mathcal{D}}_n$ is defined as in Corollary 2, and

$$N_{full}^{1/2}\big(\widehat{\eta}_{CC}^* - \widehat{\eta}_{CC}\big) = \sum_{j=1}^n W_j\frac{\widehat{\varphi}_j}{N_{full}^{1/2}}, \qquad \widehat{\varphi}_j = \big(\widehat{\varphi}_j^{(1)}, \ldots, \widehat{\varphi}_j^{(q)}\big)^\top.$$

Hence, defining

$$a_{n,j} = \left(\frac{\widehat{\psi}_j^{(1)}}{(N^{(1)})^{1/2}}, \ldots, \frac{\widehat{\psi}_j^{(q)}}{(N^{(q)})^{1/2}}, \frac{\widehat{\varphi}_j^{(1)}}{N_{full}^{1/2}}, \ldots, \frac{\widehat{\varphi}_j^{(q)}}{N_{full}^{1/2}} \right)^\top \in \mathbb{R}^{2q},$$

we can write

$$G_n^* := \begin{pmatrix} \widehat{\mathcal{D}}_n\big(\widehat{\eta}_{AC}^* - \widehat{\eta}_{AC}\big) \\ N_{full}^{1/2}\big(\widehat{\eta}_{CC}^* - \widehat{\eta}_{CC}\big) \end{pmatrix} = \sum_{j=1}^n W_j a_{n,j}.$$

Thus, conditional on $\mathcal{X}_n$, the vectors $\{W_j a_{n,j} : 1 \le j \le n\}$ form a triangular array of independent random vectors with deterministic coefficients $a_{n,j}$. From Lemma 8, it follows that $\sum_{j=1}^n a_{n,j} a_{n,j}^\top = \widehat{\Xi}_n \xrightarrow{\mathbb{P}} \tilde{V} \circ \tilde{\varrho}$, and, since q is fixed, (S43) gives $\max_{1\le j\le n} \|a_{n,j}\|^2 \xrightarrow{\mathbb{P}} 0$.

For fixed $u \in \mathbb{R}^{2q}$, let $c_{n,j} = u^\top a_{n,j}$ and $s_n^2 = \sum_{j=1}^n c_{n,j}^2 = u^\top \widehat{\Xi}_n u$. To apply the conditional Lindeberg–Feller theorem, we verify that

$$\ell_n(\varepsilon) := \frac{1}{s_n^2} \sum_{j=1}^n \mathbb{E}\left[c_{n,j}^2 W_j^2 I\{|c_{n,j} W_j| > \varepsilon s_n\} \,\middle|\, \mathcal{X}_n \right] \xrightarrow{\mathbb{P}} 0, \quad \forall \varepsilon > 0.$$

Since $s_n^2 = u^\top \widehat{\Xi}_n u \xrightarrow{\mathbb{P}} u^\top (\tilde{V} \circ \tilde{\varrho}) u$, we have $s_n^2 = O_{\mathbb{P}}(1)$, and $\max_{1\le j\le n} |c_{n,j}| \le \|u\| \max_{1\le j\le n} \|a_{n,j}\| \xrightarrow{\mathbb{P}} 0$. Hence, whenever $s_n > 0$,

$$\begin{aligned}
\ell_n(\varepsilon) &= \frac{1}{s_n^2} \sum_{j=1}^n c_{n,j}^2 \, \mathbb{E}\left[W_1^2 I\left\{ |W_1| > \frac{\varepsilon s_n}{|c_{n,j}|} \right\} \,\middle|\, \mathcal{X}_n \right] \\
&\le \frac{1}{s_n^2} \sum_{j=1}^n c_{n,j}^2 \, \mathbb{E}\left[W_1^2 I\left\{ |W_1| > \frac{\varepsilon s_n}{\max_{1\le k\le n} |c_{n,k}|} \right\} \,\middle|\, \mathcal{X}_n \right] \\
&= \mathbb{E}\left[W_1^2 I\left\{ |W_1| > \frac{\varepsilon s_n}{\max_{1\le j\le n} |c_{n,j}|} \right\} \,\middle|\, \mathcal{X}_n \right] \xrightarrow{\mathbb{P}} 0, \quad \forall \varepsilon > 0,
\end{aligned}$$

where the last convergence follows from the Dominated Convergence Theorem, since $s_n \xrightarrow{\mathbb{P}} \{u^\top (\tilde{V} \circ \tilde{\varrho}) u\}^{1/2} < \infty$, $\max_{1\le j\le n} |c_{n,j}| \xrightarrow{\mathbb{P}} 0$ and $\mathbb{E}(W_1^2) = 1$. Thus,

$$u^\top G_n^* \mid \mathcal{X}_n \xrightarrow{\mathcal{L}} \mathcal{N}\left(0, u^\top (\tilde{V} \circ \tilde{\varrho}) u\right) \quad \text{in probability.}$$

By the Cramér–Wold device,

$$G_n^* \mid \mathcal{X}_n \xrightarrow{\mathcal{L}} Z, \qquad Z \sim \mathcal{N}_{2q}\left(0, \tilde{V} \circ \tilde{\varrho}\right) \quad \text{in probability.} \tag{S50}$$

In particular, since the diagonal entries of $\widehat{\mathcal{D}}_n$ and N_{full} diverge in probability, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\|\widehat{\eta}_{AC}^* - \widehat{\eta}_{AC}\| > \varepsilon \mid \mathcal{X}_n \right) \xrightarrow{\mathbb{P}} 0, \qquad \mathbb{P}\left(\|\widehat{\eta}_{CC}^* - \widehat{\eta}_{CC}\| > \varepsilon \mid \mathcal{X}_n \right) \xrightarrow{\mathbb{P}} 0. \tag{S51}$$

Step 2: Bootstrap Wald statistics. From the Mean Value Theorem applied to each component of f, we can write

$$f(\widehat{\eta}^*_{AC}) - f(\widehat{\eta}_{AC}) = \nabla f(\tau^*_A)\,(\widehat{\eta}^*_{AC} - \widehat{\eta}_{AC}), \qquad f(\widehat{\eta}^*_{CC}) - f(\widehat{\eta}_{CC}) = \nabla f(\tau^*_C)\,(\widehat{\eta}^*_{CC} - \widehat{\eta}_{CC}),$$

where $\nabla f(\tau^*_A)$ denotes the $d \times q$ matrix whose lth row is $\nabla f^{(l)}(\tau^*_{A,l})^\top$, $\tau^*_{A,l}$ being a point of the segment joining $\widehat{\eta}_{AC}$ and $\widehat{\eta}^*_{AC}$ that depends on l, and analogously for $\nabla f(\tau^*_C)$. From the above display and using $(cA)^- = c^{-1}A^-$ for $c > 0$, we can write

$$T^*_n = \Big\{\widehat{N}^{1/2} C \nabla f(\tau^*_A) \widehat{\mathcal{D}}_n^{-1}\, \widehat{\mathcal{D}}_n\big(\widehat{\eta}^*_{AC} - \widehat{\eta}_{AC}\big)\Big\}^\top \big(\widehat{N}\, C \widehat{\Omega}_f C^\top\big)^- \Big\{\widehat{N}^{1/2} C \nabla f(\tau^*_A) \widehat{\mathcal{D}}_n^{-1}\, \widehat{\mathcal{D}}_n\big(\widehat{\eta}^*_{AC} - \widehat{\eta}_{AC}\big)\Big\},$$

and

$$T^{CC*}_n = \Big\{q^{1/2} C \nabla f(\tau^*_C)\, N_{full}^{1/2}\big(\widehat{\eta}^*_{CC} - \widehat{\eta}_{CC}\big)\Big\}^\top \big(q N_{full}\, C \widehat{\Omega}^{CC}_f C^\top\big)^- \Big\{q^{1/2} C \nabla f(\tau^*_C)\, N_{full}^{1/2}\big(\widehat{\eta}^*_{CC} - \widehat{\eta}_{CC}\big)\Big\}.$$

Note that $\widehat{N}^{1/2}\widehat{\mathcal{D}}_n^{-1} \xrightarrow{\mathbb{P}} K$, where K is as defined in Corollary 3. From Proposition 5, we have that

$$\widehat{N}\, C \widehat{\Omega}_f C^\top \xrightarrow{\mathbb{P}} \Sigma_f, \qquad q N_{full}\, C \widehat{\Omega}^{CC}_f C^\top \xrightarrow{\mathbb{P}} \Sigma^{CC}_f = q C \nabla f(\eta) V \nabla f(\eta)^\top C^\top,$$

both limits being positive definite. We have that $\|\tau^*_{A,l} - \eta\| \le \|\widehat{\eta}_{AC} - \eta\| + \|\widehat{\eta}^*_{AC} - \widehat{\eta}_{AC}\|$. Now from Theorem 1, (S51) and the continuity of the partial derivatives at η give, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\|\nabla f(\tau^*_A) - \nabla f(\eta)\| > \varepsilon \mid \mathcal{X}_n\right) \xrightarrow{\mathbb{P}} 0, \qquad \mathbb{P}\left(\|\nabla f(\tau^*_C) - \nabla f(\eta)\| > \varepsilon \mid \mathcal{X}_n\right) \xrightarrow{\mathbb{P}} 0. \tag{S52}$$

Let Z as in (S50) and write $Z = (Z_1^\top, Z_2^\top)^\top$, $Z_i \in \mathbb{R}^q$, $i = 1, 2$. All these convergences imply that we can write

$$T^*_n = \Big(C \nabla f(\eta) K\, Z_1\Big)^\top \Sigma_f^- \Big(C \nabla f(\eta) K\, Z_1\Big) + r_1,$$

and

$$T^{CC*}_n = \Big(q^{1/2} C \nabla f(\eta)\, Z_2\Big)^\top \big(\Sigma^{CC}_f\big)^- \Big(q^{1/2} C \nabla f(\eta)\, Z_2\Big) + r_2.$$

with $\mathbb{P}\left(\|r_1\| > \varepsilon \mid \mathcal{X}_n\right) \xrightarrow{\mathbb{P}} 0$ and $\mathbb{P}\left(\|r_2\| > \varepsilon \mid \mathcal{X}_n\right) \xrightarrow{\mathbb{P}} 0$ for every $\varepsilon > 0$. This fact and (S50) give that

$$\big(T^*_n, T^{CC*}_n\big) \mid \mathcal{X}_n \xrightarrow{\mathcal{L}} (\chi_A, \chi_C) \quad \text{in probability}$$

where (χ_A, χ_C) is the limiting distribution of (T_n, T^{CC}_n). Therefore, applying the Continuous Mapping Theorem, we get that $F^*_n(t) \xrightarrow{\mathbb{P}} F_0(t)$ at every continuity point of F_0.

Finally, χ_A and χ_C have continuous χ^2_c marginal distributions, so that, for every $t \in \mathbb{R}$,

$$\mathbb{P}\{\max(\chi_A, \chi_C) = t\} \leq \mathbb{P}(\chi_A = t) + \mathbb{P}(\chi_C = t) = 0,$$

their maximum has no point masses and F_0 is continuous. Equation (23) then follows from Pólya's Theorem.

Step 3: Contiguous and fixed alternatives Suppose first that $N^{1/2}Cf(\eta) \to b \in \mathbb{R}^c$. Write

$$\widehat{N}^{1/2}Cf(\widehat{\eta}_{AC}) = \widehat{N}^{1/2}C\{f(\widehat{\eta}_{AC}) - f(\eta)\} + (\widehat{N}/N)^{1/2}N^{1/2}Cf(\eta),$$

$$(qN_{\text{full}})^{1/2}Cf(\widehat{\eta}_{CC}) = (qN_{\text{full}})^{1/2}C\{f(\widehat{\eta}_{CC}) - f(\eta)\} + (qN_{\text{full}}/N)^{1/2}N^{1/2}Cf(\eta).$$

Since $\widehat{N}/N \xrightarrow{\mathbb{P}} 1$ (see the proof of Proposition 5) and $N_{\text{full}}/(n\pi_n^{full}) \xrightarrow{\mathbb{P}} 1$ (see the proof of Lemma 8), which together with Assumption A9 gives $N_{\text{full}}/N \xrightarrow{\mathbb{P}} \kappa_{\text{full}}$, the second terms on the right-hand sides converge in probability to b and $(q\kappa_{\text{full}})^{1/2}b$, respectively. As for the first terms, the Mean Value Theorem, Theorem 1, the continuity of the partial derivatives of f at η and $\widehat{N}^{1/2}\widehat{\mathcal{D}}_n^{-1} \xrightarrow{\mathbb{P}} K$, as in Step 2, together with Lemma 1, give

$$\Big[\widehat{N}^{1/2}C\{f(\widehat{\eta}_{AC}) - f(\eta)\},\, (qN_{\text{full}})^{1/2}C\{f(\widehat{\eta}_{CC}) - f(\eta)\}\Big]^\top \xrightarrow{\mathcal{L}} (Y_A^\top, Y_C^\top)^\top,$$

where $Y_A = C\nabla f(\eta)KZ_1$ and $Y_C = q^{1/2}C\nabla f(\eta)Z_2$, with $(Z_1^\top, Z_2^\top)^\top \sim \mathcal{N}_{2q}(0, \tilde{V} \circ \tilde{\varrho})$. Hence, by Slutsky's Theorem,

$$\big\{\widehat{N}^{1/2}Cf(\widehat{\eta}_{AC}),\, (qN_{\text{full}})^{1/2}Cf(\widehat{\eta}_{CC})\big\}^\top \xrightarrow{\mathcal{L}} \big\{Y_A + b,\, Y_C + (q\kappa_{\text{full}})^{1/2}b\big\}^\top.$$

Now, using $(cA)^- = c^{-1}A^-$ for $c > 0$ as in Step 2,

$$T_n = \{\widehat{N}^{1/2}Cf(\widehat{\eta}_{AC})\}^\top \big(\widehat{N}\, C\widehat{\Omega}_f C^\top\big)^- \{\widehat{N}^{1/2}Cf(\widehat{\eta}_{AC})\},$$

$$T_n^{CC} = \{(qN_{\text{full}})^{1/2}Cf(\widehat{\eta}_{CC})\}^\top \big(qN_{\text{full}}\, C\widehat{\Omega}_f^{CC} C^\top\big)^- \{(qN_{\text{full}})^{1/2}Cf(\widehat{\eta}_{CC})\},$$

and, from Proposition 5, $\widehat{N}\, C\widehat{\Omega}_f C^\top \xrightarrow{\mathbb{P}} \Sigma_f$ and $qN_{\text{full}}\, C\widehat{\Omega}_f^{CC} C^\top \xrightarrow{\mathbb{P}} \Sigma_f^{CC}$, both positive definite, so that their generalized inverses converge in probability to Σ_f^{-1} and $(\Sigma_f^{CC})^{-1}$. The Continuous Mapping Theorem then gives

$$(T_n, T_n^{CC}) \xrightarrow{\mathcal{L}} (\chi_{A,b}, \chi_{C,b}),$$

where $\chi_{A,b} = (Y_A + b)^\top \Sigma_f^{-1}(Y_A + b)$ and $\chi_{C,b} = \big\{Y_C + (q\kappa_{\text{full}})^{1/2}b\big\}^\top (\Sigma_f^{CC})^{-1}\big\{Y_C + (q\kappa_{\text{full}})^{1/2}b\big\}$, and, again by the Continuous Mapping Theorem, $T_n^{\max} \xrightarrow{\mathcal{L}} \max(\chi_{A,b}, \chi_{C,b})$. Let $F_b(t) = \mathbb{P}\{\max(\chi_{A,b}, \chi_{C,b}) \leq t\}$. Since Σ_f and Σ_f^{CC} are the covariance matrices of Y_A and Y_C, $\chi_{A,b}$ and $\chi_{C,b}$ have noncentral χ^2_c distributions, which are continuous, and the

argument of Step 2 shows that F_b is continuous. Notice that for $b = 0$ these are the limits obtained in Step 2, so that F_0 is the distribution function of $m(Y)$, where $Y = (Y_A^\top, Y_C^\top)^\top$ and

$$m(y) = \max\left\{y_A^\top \Sigma_f^{-1} y_A,\ y_C^\top (\Sigma_f^{CC})^{-1} y_C\right\}, \qquad y = (y_A^\top, y_C^\top)^\top \in \mathbb{R}^{2c}.$$

Now, note that (23) holds whatever the value of $Cf(\eta)$. Moreover, F_0 is continuous (Step 2) and strictly increasing on $(0, \infty)$: the support of the centered Gaussian vector Y is a linear subspace, and m is continuous, positive off the origin, because Σ_f and Σ_f^{CC} are positive definite, and satisfies $m(\lambda y) = \lambda^2 m(y)$ for $\lambda > 0$, so that, for $0 < t < t'$, the open set $\{y : t < m(y) < t'\}$ meets the support of Y and $F_0(t') - F_0(t) = \mathbb{P}\{t < m(Y) \leq t'\} > 0$. Therefore, the $(1-\alpha)$-quantile $c_{1-\alpha}$ of F_0 is unique and, from (23), the $(1-\alpha)$-quantile $c^*_{1-\alpha}$ of F_n^* satisfies $c^*_{1-\alpha} \xrightarrow{\mathbb{P}} c_{1-\alpha}$. From $T_n^{\max} \xrightarrow{\mathcal{L}} \max(\chi_{A,b}, \chi_{C,b})$ and $c^*_{1-\alpha} \xrightarrow{\mathbb{P}} c_{1-\alpha}$, Slutsky's Theorem yields $T_n^{\max} - c^*_{1-\alpha} \xrightarrow{\mathcal{L}} \max(\chi_{A,b}, \chi_{C,b}) - c_{1-\alpha}$. Since F_b is continuous, $\mathbb{P}(T_n^{\max} > c^*_{1-\alpha}) \to 1 - F_b(c_{1-\alpha}) =: \beta(b)$. For $b = 0$, $F_b = F_0$ and $F_0(c_{1-\alpha}) = 1 - \alpha$, so $\beta(0) = \alpha$, proving part 1.

For part 2, it suffices to show $F_b(t) \leq F_0(t)$ for all $t > 0$. Let $a = \{b^\top, (q\kappa_{\text{full}})^{1/2} b^\top\}^\top$ and $A_t = \{(y_A^\top, y_C^\top)^\top : y_A^\top \Sigma_f^{-1} y_A \leq t,\ y_C^\top (\Sigma_f^{CC})^{-1} y_C \leq t\}$. Then $F_b(t) = \mathbb{P}(Y + a \in A_t)$ and $F_0(t) = \mathbb{P}(Y \in A_t)$. Since A_t is closed, convex and symmetric, Anderson's Theorem (see e.g., Theorem 4.2.4 in Tong (1990)) would give the result if $\mathbb{C}\text{ov}(Y)$ were positive definite. To cover the possibly singular case , let $G \sim \mathcal{N}_{2c}(0, I_{2c})$ be independent of Y and set $Y_\varepsilon = Y + \varepsilon G$. Then $\mathbb{C}\text{ov}(Y_\varepsilon)$ is positive definite and hence $\mathbb{P}(Y_\varepsilon + a \in A_t) \leq \mathbb{P}(Y_\varepsilon \in A_t)$. Since $Y_\varepsilon \xrightarrow{a.s.} Y$ and $Y_\varepsilon + a \xrightarrow{a.s.} Y + a$, and $\mathbb{P}(Y \in \partial A_t) = \mathbb{P}(Y + a \in \partial A_t) = 0$, we have that $\mathbf{1}_{\{Y_\varepsilon \in A_t\}} \xrightarrow{a.s.} \mathbf{1}_{\{Y \in A_t\}}$, and $\mathbf{1}_{\{Y_\varepsilon + a \in A_t\}} \xrightarrow{a.s.} \mathbf{1}_{\{Y + a \in A_t\}}$. Since these indicators are bounded by 1, the dominated convergence theorem gives $\mathbb{P}(Y_\varepsilon \in A_t) \to F_0(t)$ and $\mathbb{P}(Y_\varepsilon + a \in A_t) \to F_b(t)$.

Finally, if $\|N^{1/2} Cf(\eta)\| \to \infty$, part 2 of Proposition 5 gives $T_n \xrightarrow{\mathbb{P}} \infty$. As shown above, (23) also holds in this case, so that $c^*_{1-\alpha} \xrightarrow{\mathbb{P}} c_{1-\alpha} < \infty$. Since $T_n^{\max} \geq T_n$, for any $M > c_{1-\alpha}$,

$$\mathbb{P}(T_n^{\max} > c^*_{1-\alpha}) \geq \mathbb{P}(T_n > M) - \mathbb{P}(c^*_{1-\alpha} > M) \to 1,$$

which proves part 3. □

Proof of Proposition 4. The proof follows the same argument as that of Theorem 3, applied population by population. □

S2 Classical results on U-statistics

In this section, we present some properties of U-statistics, which will be instrumental in proving the main results later in the paper. The results and definitions not explicitly

referenced can be found in Sections 5.1 and 5.2 of Serfling (1980).

Let $h(x_1, \ldots, x_m)$ be a symmetric kernel of degree m. Let $X_1, \ldots, X_n$ be a random sample with $n \geq m$. The U-statistic based on h and the random sample $X_1, \ldots, X_n$ is defined as

$$U_n = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \cdots < i_m \leq n} h(X_{i_1}, \ldots, X_{i_m}).$$

We say that U_n is a degree-m U-statistic. Let $\theta = \mathbb{E}\{h(X_1, \ldots, X_m)\}$. The U-statistic U_n is a symmetric function of its inputs and serves as an unbiased estimator of θ, i.e.,

$$\mathbb{E}(U_n) = \theta.$$

To compute the variance of U_n, we first define the conditional expectation of the kernel h given c variables

$$h_c(x_1, \ldots, x_c) = \mathbb{E}\{h(X_1, \ldots, X_m) \mid X_1 = x_1, \ldots, X_c = x_c\}. \tag{S53}$$

We define the centered version of h_c as

$$\widetilde{h}_c(x_1, \ldots, x_c) = h_c(x_1, \ldots, x_c) - \theta. \tag{S54}$$

Note that, $\mathbb{E}[h_c(X_1, \ldots, X_c)] = \theta$. The variance of both h_c and $\widetilde{h}_c$ is

$$\sigma_c^2 = \mathbb{V}\{h_c(X_1, \ldots, X_c)\} = \mathbb{V}\left\{\widetilde{h}_c(X_1, \ldots, X_c)\right\}, \quad c = 1, \ldots, m. \tag{S55}$$

We also have that for indices $i_1 < \cdots < i_r$ and $j_1 < \cdots < j_s$ sharing exactly c common elements, $\#\left(\{i_1, \ldots, i_r\} \cap \{j_1, \ldots, j_s\}\right) = c$, where $\#(S)$ denotes the cardinality of S,

$$\sigma_c^2 = \mathbb{C}\mathrm{ov}\left\{h_r(X_{i_1}, \ldots, X_{i_r}), h_s(X_{j_1}, \ldots, X_{j_s})\right\}. \tag{S56}$$

Note that the previous equation is valid if we replace the kernels for their centered version as well. The variances σ_c^2 satisfy the ordering,

$$\sigma_1^2 \leq \sigma_2^2 \leq \cdots \leq \sigma_m^2. \tag{S57}$$

Using these variances, the variance of U_n is given by

$$\mathbb{V}(U_n) = \binom{n}{m}^{-1} \sum_{c=1}^{m} \binom{m}{c} \binom{n-m}{m-c} \sigma_c^2. \tag{S58}$$

S3 Gaussian copula missingness mechanism

A Gaussian copula couples arbitrary marginal distributions through the correlation structure of a latent Gaussian vector. Here it is used to make the components of M_j dependent while preserving Assumption A3. For each observation $1 \le j \le n$, consider a latent vector

$$W_j \sim \mathcal{N}_p\big(0, (1-\theta)I_4 + \theta J_4\big),$$

where θ is the common correlation between the latent variables, and define the random availability probabilities by

$$\varpi_j^{(\ell)} = \frac{1}{2}\Big\{1 + \Phi\big(W_j^{(\ell)}\big)\Big\} \sim U(1/2, 1)\,, \qquad 1 \le \ell \le p.$$

Given ϖ_j, the indicators $M_j^{(\ell)} \sim \text{Bernoulli}(\varpi_j^{(\ell)})$ are conditionally independent. Since (ϖ_j, M_j) is independent of X_j, the mechanism is MCAR. Moreover, the latent correlation matrix is positive semidefinite whenever $\theta > -1/(p-1)$.

If $A_r = \{r\}$, then, the availability quantities appearing in the theory admit closed-form expressions as functions of θ. Since in this experiment,

$$\pi^{(r)} = \mathbb{P}\left(R_j^{(r)} = 1\right) = \mathbb{P}\left(M_j^{(r)} = 1\right) = \mathbb{E}\Big\{\mathbb{P}\left(M^{(r)} = 1 \,\Big|\, \varpi_j^{(r)}\right)\Big\} = \mathbb{E}\left(\varpi_j^{(r)}\right) = 3/4,$$

and

$$\begin{aligned}
\pi^{(r,s)} &= \mathbb{P}\left(R_j^{(r)} = R_j^{(s)} = 1\right) = \mathbb{P}\left(M_j^{(r)} = M_j^{(s)} = 1\right) \\
&= \mathbb{E}\Big\{\mathbb{P}\Big(M_j^{(r)} = M_j^{(s)} = 1 \,\Big|\, \varpi_j^{(r)}, \varpi_j^{(s)}\Big)\Big\} = \mathbb{E}\Big(\varpi_j^{(r)}\varpi_j^{(s)}\Big) \\
&= \frac{1}{2} + \frac{1}{4}\,\mathbb{E}\Big\{\Phi\Big(W_j^{(r)}\Big)\,\Phi\Big(W_j^{(s)}\Big)\Big\} \\
&= \frac{9}{16} + \frac{\arcsin(\theta/2)}{8\pi},
\end{aligned}$$

for $r \neq s$. The penultimate equality follows by introducing two independent standard normal random variables Z_1 and Z_2, independent of $(W_j^{(r)}, W_j^{(s)})$, so that

$$\mathbb{E}\Big\{\Phi\left(W_j^{(r)}\right)\Phi\left(W_j^{(s)}\right)\Big\} = \mathbb{P}\Big(W_j^{(r)} - Z_1 > 0,\, W_j^{(s)} - Z_2 > 0\Big)\,,$$

and then applying the classical formula for the bivariate Gaussian orthant probability (see, e.g., equation (8.2.6) of Tong (1990)).

S4 Average sample sizes in the simulation study of Subsection 8.1 of the paper

Table S1: Average sample sizes in the size study of Table 2. For each q, $\bar{N}_{AC}$ is the average of the mean per-component available sample sizes among the q components involved, and $\bar{N}_{CC}$ is the average CC sample size.

Case	n	$\rho_x = 0.3$						$\rho_x = 0.95$					
		$q=2$		$q=3$		$q=4$		$q=2$		$q=3$		$q=4$	
		$\bar{N}_{AC}$	$\bar{N}_{CC}$	$\bar{N}_{AC}$	$\bar{N}_{CC}$	$\bar{N}_{AC}$	$\bar{N}_{CC}$	$\bar{N}_{AC}$	$\bar{N}_{CC}$	$\bar{N}_{AC}$	$\bar{N}_{CC}$	$\bar{N}_{AC}$	$\bar{N}_{CC}$
1.	50	50	50	50	50	50	50	50	50	50	50	50	50
	100	100	100	100	100	100	100	100	100	100	100	100	100
	250	250	250	250	250	250	250	250	250	250	250	250	250
	500	500	500	500	500	500	500	500	500	500	500	500	500
2.	50	38	28	37	21	38	16	38	28	38	21	38	16
	100	75	56	75	42	75	32	75	56	75	42	75	32
	250	188	141	188	106	188	79	187	141	188	106	188	79
	500	375	281	375	211	375	158	375	281	375	211	375	158
3.	50	42	36	40	25	38	15	42	36	40	25	37	15
	100	85	72	80	50	75	30	85	72	80	50	75	30
	250	213	180	200	126	187	76	212	180	200	126	187	75
	500	425	360	400	252	375	151	425	360	400	252	375	151
4.	50	38	28	38	22	37	17	38	28	38	22	38	17
	100	75	57	75	43	75	34	75	57	75	44	75	34
	250	188	142	188	109	188	84	187	142	187	109	187	84
	500	375	284	375	218	375	168	375	284	375	218	375	168
5.	50	37	28	37	21	37	16	38	28	37	21	38	16
	100	75	56	75	42	75	32	75	56	75	42	75	32
	250	188	141	187	106	188	79	188	141	188	105	188	79
	500	375	281	375	211	375	158	375	281	375	211	375	158
6.	50	38	28	38	20	38	15	38	28	38	20	38	15
	100	75	56	75	41	75	30	75	56	75	41	75	30
	250	188	139	188	102	187	74	188	139	187	102	187	74
	500	375	278	375	204	375	148	375	278	375	204	375	148

References

Graybill, F. A. (1976). *Theory and Application of the Linear Model*. Duxbury Press, North Scituate, Massachusetts.

Horn, R. and Johnson, C. (2012). *Matrix Analysis*. Cambridge University Press.

Romero-Madroñal, M., Sillero-Denamiel, M., and Jiménez-Gamero, M. D. (2026). Testing the equality of estimable parameters. *Preprint arXiv:2607.07588*.

Serfling, R. J. (1980). *Approximation Theorems of Mathematical Statistics*. Wiley, New York.

Tong, Y. L. (1990). *The Multivariate Normal Distribution*. Springer Series in Statistics. Springer-Verlag, New York.